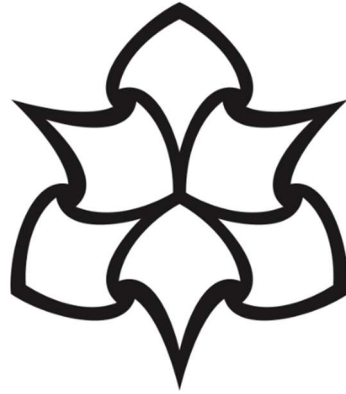




**Manchester
Metropolitan
University**

**A web game to measure risk-taking and
collect results, for use in psychological
experiments.**

A DISSERTATION SUBMITTED
TO MANCHESTER METROPOLITAN UNIVERSITY

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Declaration

No part of this project has been submitted in support of an application for any other degree or qualification at this or any other institute of learning. Apart from those parts of the project containing citations to the work of others, this project is my own unaided work. This work has been carried out in accordance with the Manchester Metropolitan University research ethics procedures, and has received ethical approval number 76392.

Signed: Ahsan Siddiqui

Date: 29/04/2025

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Abstract

This paper presents the development of a novel risk-taking assessment tool, made for use in psychological research. This project attempts to create a system that can effectively measure risk in individuals as well as produce the necessary data for researchers to use in experiments. Extensive research has gone into the design of this product and improving its systems to ensure academic and professional validity

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Chapter 1 – Introduction

1.1 - Background

In psychology, risk-taking behaviour is a central area of study, affecting decision making, social behaviour, interpersonal behaviour and intrinsic motivation (Steinberg, 2010). Traditional experiments such as the Balloon Analogue Risk Task (BART) have been widely utilised to measure and quantify risk-taking behaviour in controlled environments (Lejuez et al., 2002). Likewise, the Iowa Gambling Task (IGT) has significantly contributed to this specific field of study in psychology, providing valuable insight into what motivates people to make certain decisions (Bechara et al., 1994). Despite their significant contributions to this field, both models lack in some aspects, which severely limit their experimental validity.

The IGT has been critiqued on the fact that past scientific studies have several gaps in their analysis, specifically there is a lack of expert knowledge and comprehensive analysis pertaining to the mental processes that directly influence and affect the decision-making process (Burdick, Roy and Raver, 2013). Similarly, with the BART, a major drawback to the study is that many of its models assume that participants make decisions based on a constant and given probability of negative outcomes which can be considered a generalisation. (Zhou et al., 2021). Moreover, many of these experiments are based on repetitive or abstract task designs which could potentially limit user engagement which would directly affect the validity of results.

More recent studies have suggested that gamification and motivational design can be extremely beneficial in enhancing user motivation and data reliability, especially when pertaining to behavioural research (Hamari, Koivisto and Sarsa, 2014). Gamification has been used in the past in behavioural experiments which demonstrates its validity and feasibility for use in behavioural experiments (Knežević and Jovanović, 2021). Prospect theory also highlights that individuals may overweigh a loss more than a winning of the same proportion, this suggests risk-taking is highly context-sensitive and psychologically nuanced (Kahneman and Tversky, 1979).

The critical limitation in this specific area of study in psychology is the inability to link all these studies together and utilise more recent studies to improve on the already existing foundation. An example of this limitation is that these studies fail to gauge both qualitative and quantitative data simultaneously, often failing to capture the participants' internal reasoning and motivations behind the decisions that they make. (Cyders and Coskunpinar, 2011). Utilising data directly from the participants could prove useful to understanding what motivates individuals making decisions, specifically decisions that would be hard to understand through raw numbers.

In response to these limitations, this project will be designed to address the flaws of the previous models in this field while also building on the already established framework. This

project proposes to design and gamify a risk-taking experiment which will be available to play through a web browser.

1.2 - Aims

The aim of this project is to create a web-game to measure the willingness to take risks within participants. The gameplay data will be collected and used by psychologists to conduct research and be used in psychological experiments.

1.3 - Objectives

The following are smaller objectives in the project that fall into the greater aim for the task. They are specific criteria that will be completed as part of the main aim for the project.

- Develop a game that can test a user's ability and willingness to take risks.
- Collect meaningful and anonymous data from the user's gameplay store it correctly for researchers to use.
- Make the game as accessible as possible, catering to all age groups and backgrounds and minimising any gameplay features that could cause discomfort or stress.
- Collect both qualitative and quantitative data from participants through surveys to understand the motivations behind decisions.

1.4 - Report Organisation

This report is broken down into six distinct chapters, all tackling a different aspect of the project.

1.4.1 – Chapter 2 - Literature Review

This chapter critically reviews existing literature and study related to risk-taking and gamification in the field of behavioural research in models such as IGT or the BART. Gaps in current models and approaches are highlighted while establishing the academic base for this project.

1.4.2 – Chapter 3 – Design

This chapter details the design process of this game, discussing the motivation and rationale behind design choices such as the user interface, accessibility considerations, gameplay mechanics and how each of the features address the project's research goals.

1.4.3 – Chapter 4 - Implementation

This chapter discusses the technologies, coding tools and technical approach to create this game. It explores how the game was deployed for experimental use.

1.4.4 – Chapter 5 – Analysis

This chapter will evaluate the effectiveness of the project's intended use based on user testing, the collection of data and feedback from participants. This area discusses if the

project met its aims and objectives as well as how users responded to data collection and gameplay mechanics.

1.4.5 – Chapter 6 – Conclusion

The concluding chapter of this report will summarise the reflect on the overall project, acknowledges limitations and will propose areas for future improvement and work.

Chapter 2 – Literature review

2.1 - Psychological Models of Risk-Taking

In psychology, risk-taking behaviour is categorised as the tendency for individuals to partake in actions that could expose them to the possibility of negative consequences or outcomes. This behaviour typically occurs in pursuit of a favourable outcome or a reward. This behaviour plays a critical role in fields such as behavioural economics, developmental psychology, and social psychology. Risk-taking behaviour can aid researchers in understanding why and how individuals make potentially harmful choices in a wide variety of situations (Steinberg, 2010).

An example of this understanding being used is the “Near-miss effect”. This effect refers to a situation where an individual recognises a loss as an “almost win”. This activity in the individual can activate reward-related areas in the brain which reinforce the behaviour despite its negative outcome which subsequently leads the individual to have an increase in motivation (Clark et al., 2009). However, the near miss effect may not be experienced uniformly across individuals. More recent research has indicated that personal susceptibility to a gambling addiction can significantly influence the degree to which the near-miss experiences can affect motivation. This suggests that the effect is influenced by individual differences (Habib & Dixon, 2010). Understanding this nuance is essential in creating a risk assessment tool that accounts for variability in individual responses to near-miss events.

This also links with the “Illusion of control” which is the tendency for individuals to overestimate their ability to control events or outcomes, this bias is particularly prevalent in games of chance. This cognitive bias can lead to the individual having increased gambling behaviour as they believe that their actions influence their outcomes. (Langer, 1975). Nonetheless, this concept has been critiqued for not completely accounting for motivational or emotional factors. These studies also suggest that the desire for control or predictability, particularly in uncertain situations, can lead to the persistence of a cognitive bias, regardless of if the users are aware of the randomness involved (Thompson et al., 1998). Comprehending this flaw in the illusion of control is important for developing interventions that efficiently assess the underlying motivations behind decision making.

One influential theory is Steinberg’s Dual Systems Model. The dual systems model indicates that risk-taking behaviour is the product of the brain’s imbalance between the sensitivity system and cognitive control system (Steinberg, 2010). The dual systems model explains why risk-taking behaviour is more prevalent during adolescence, however it also provides necessary insight into the decision-making process in adults, especially in emotionally charged or reward driven situations. Overall, this theory defines understanding of risk-taking in development psychology and how the behaviour varies with age. Despite this, the dual systems model has faced criticism for oversimplifying adolescent risk-taking behaviour. Critics argue that it does not completely consider the influence of other factors such as environmental or cultural factors. These variables are crucial in forming the decision-making process (Casey & Jones, 2010). This limitation suggests that while the model provides a

beneficial framework, it may not fully capture the nuances and complexities of risk-taking behaviour, especially in real world contexts which are inherently diverse. This limitation is particularly relevant for this project which aims to adhere to a wide range of demographics.

Additionally, prospect theory is a foundational study which suggests that individuals do not evaluate potential gains and losses through an objective lens, instead they evaluate gains and losses from a subjective lens which emphasises loss aversion and a partial probability weighting (Kahneman and Tversky, 1979). This theory highlights why individuals may feel opposed to making risky choices even when the expected outcome is positive. Similarly, it also explains the motivation behind high-risk behaviour in the pursuit of avoiding losses. While prospect theory has significantly furthered the understanding of decision-making under risk, it has been critiqued for its limited cross-culture adaptability. Research indicates that risk-taking preferences can differ heavily across different regional, cultural or social contexts. This suggests that the theory may not apply universally for risk-taking behaviour in individuals (Weber & Hsee, 1998). This limitation is especially applicable for this project, which aims to develop a risk-assessment tool which is appropriate for diverse populations.

Furthermore, risk-taking behaviour is influenced by a range of factors including personality traits, age, and emotional state (Zuckerman, 2007). These variables complicate the process of designing risk-measurement tools as no single model can account for all behavioural factors and variables. While these models emphasise the importance of this field of study, they also act as the base and the theoretical foundation for many risk-assessment tools which will be reviewed in the following section.

2.2 - Existing Risk-Measurement Tools

2.2.1 – Balloon Analogue Risk Task

The Balloon Analogue Risk Task (BART) is a risk assessment tool developed by Lejuez et al. (2002). It is a computerized behavioural measuring tool which was developed to test risk-taking tendencies in individuals. In the task, users are displayed a digital balloon which will inflate incrementally when prompted by the user. Every time the user pumps the balloon, the potential monetary gain increases, but so does the risk of the balloon bursting, which would result in the user losing all of their accumulated money from that specific trial. Participants can also choose to “cash out” which would result in them taking away all the money from that specific trial. The index for risk-taking behaviour is the average number of pumps on unexploded balloons.

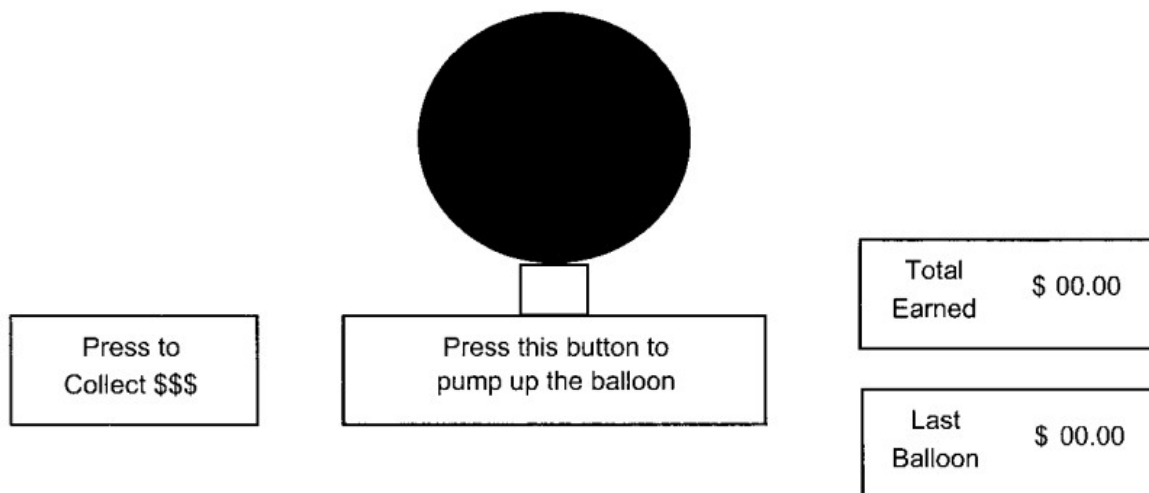


Figure 1 - A diagram of the Balloon Analogue Risk Task (Lejuez et al. 2002)

The BART has been applied widely across several domains. The research has been applied to Substance use research where BART scores have been linked to real world risk-taking behaviours, including alcohol and drug use. This suggests its viability in identifying individuals with higher risk-taking tendencies (Lejuez et al., 2002; Fernie et al., 2010). Likewise, the BART has also been employed to study impulsivity and decision-making in individuals with ADHD, Schizophrenia and other psychological conditions (Hunt et al., 2005). This suggests the model's viability when studying neurodivergent populations which adds a level of inclusivity, allowing the study to reach a larger range of individuals. Additionally, BART has been applied towards adolescent studies, helping examine and understand developmental trajectories with risk-taking behaviours. Its use in adolescent studies suggests that this study is appropriate for studying this behaviour in younger populations, which would increase the range of individuals that this experiment can apply to.

Despite its widespread uses, BART has faced several criticisms. The first of many is pertaining to the task's ecological validity. The task's artificial setting may not accurately reflect real-world risk-taking behaviour or scenarios which limits its ecological validity highly (Reynolds et al., 2006). Additionally, the model falls victim to "learning effects". This is the idea that participants may learn the task's contingencies over time which could result in the behaviour of participants to be influenced or deviate from how they would normally act in a risk-taking environment (Pleskac et al., 2008). Naturally, this reduces the experimental validity of the model as the model fails to adapt to individuals. Furthermore, BART may not fully account for individual differences in risk-perception or decision-making strategies which could potentially lead to results being misinterpreted. This flaw is also a result of qualitative data not being collected which studies the reasoning behind decisions being made. On top of this, task variability is not taken into account with the BART. There are many variation in the task's parameters such as explosion probability, reward structure and the number of balloons. The variation of these parameters across studies can affect comparability

and generalisation of findings (White et al., 2008). This flaw may be a result of the task's inability to pander widely to a larger variety of audiences which results in researchers needing to modify certain aspects of the model to better fit their specific research.

Nevertheless, many modifications could be made to the BART model to address its limitations and aid in developing a model which could expand from its downsides. The first limitation to address is the linear gameplay of the model. Adaptive algorithms could be implemented to adjust explosion probabilities based on participant behaviour. This could mitigate the effect of learning effects and better capture individual risk-taking tendencies (Wallsten et al., 2005). The BART could also benefit from implementing more realistic and contextually relevant scenarios, this could improve the ecological validity of the experiment which was mentioned earlier, as well as improve participant engagement (Figner et al., 2009). Finally, BART could apply standardisation protocols across its different implementations which would increase comparability across studies and facilitate meta-analyses (Lejuez et al., 2003).

2.2.2 – Iowa Gambling task

The Iowa Gambling Task is a risk assessment tool developed by Bechara et al. (1994). It is a widely used neuropsychological test which is designed to simulate real-life risk-taking and decision-making under uncertainty. In the task, participants are presented with 4 decks of cards, decks A, B, C and D. Each card selection results in either a gain or loss in virtual money. Two decks (A and B) are considered disadvantageous, offering high short-term rewards but giving out larger long-term losses. The other 2 decks (C and D) are considered advantageous, giving smaller short-term rewards but granting higher gain over time. The IGT measures the participant's ability to learn from their behaviour and adapt their strategy to maximise earnings. This trait of the assessment makes it particularly useful when studying real world decision-making and impulsivity.

	'Disadvantageous' decks		'Advantageous' decks	
	A	B	C	D
Gain per card	100	100	50	50
Loss per 10 cards	1250	1250	250	250
Net per 10 cards	-250	-250	+250	+250

Figure 2 - Schematic diagram of the Iowa Gambling Task (Gescheidt et al., 2012)

The IGT has found many applications across several psychological domains. The first of many applications is its use in neuropsychology. The test was originally designed to assess patients with ventromedial prefrontal cortex (vmPFC) damage, who demonstrated difficulty in making decisions in their personal and financial lives, despite retaining general intelligence (Bechara et al., 1994). The IGT has also been used to study populations with substance abuse, gambling, addiction, OCD and ADHD, where impaired long-term decision making is often observed (Buelow and Suhr, 2009). Finally, the IGT has found applications in gender differences as well. Some studies have explored gender-based differences in IGT performances with inconsistent findings. This suggests that cultural and individual factors might mediate such effects (van den Bos, Homberg and de Visser, 2013).

While the IGT is one of the most used tools used for studying decision-making, it has drawn significant criticism. The first reason is its ambiguity and complexity. The task involves complex contingencies, which makes it hard to determine whether a participant is failing due to impaired decision making, confusion or poor learning (Dunn, Dalgleish and Lawrence, 2006). Another weakness of the study is the "Deck B phenomenon". Several studies show that participants repeatedly pick Deck B (Disadvantageous in the long term but presenting high immediate rewards), challenging the assumption that individuals naturally shift toward long-term gain strategies (Chiu and Lin, 2007). The IGT has also been criticised because there have been recent studies that challenge the somatic marker hypothesis. The IGT was designed to support Damasio's Somatic Marker Hypothesis, which claims that emotional bodily responses guide decision-making in individuals. However, other studies have shown

that participants may make the correct decisions without emotional cues, which casts doubts on the theory's exclusivity (Maia and McClelland, 2004).

To address its many criticisms, many suggestions and variations can be made to the IGT in the pursuit of a possible improvement. The first suggestion is simplified or modified versions. Tasks like the "Soochow Gambling task" have been developed to reduce the influence of immediate reward frequency and isolate long-term decision strategies (Chiu et al., 2008). The IGT could also benefit from cognitive modelling. This would include incorporating computational models of decision-making which can help differentiate between strategy learning and actual risk preference. This makes the results more interpretable (Busemeyer and Stout, 2002). The test could also be adapted to different cultures, testing its assumptions across them to allow for more accurate results.

2.2.3 – Columbia Card Task

The Columbia Card Task (CCT) is a behavioural measure designed by (Figner et al. 2009) to assess risk-taking behaviour under varying emotional and cognitive conditions. The task mainly exists in two versions. The first version is Hot CCT in which participants make decision incrementally while receiving immediate feedback every time they turn a card. This engages affective processes. The second version is Cold CCT in which participants decide in advance how many cards they wish to turn over. In this version, feedback is only provided at the end which engages in deliberative processes. In both versions, participants are presented with a grid of facedown cards. Every time a card is turned over, it either results in a gain or a loss, with the number of gains and losses as well as probabilities being systematically manipulated across trials

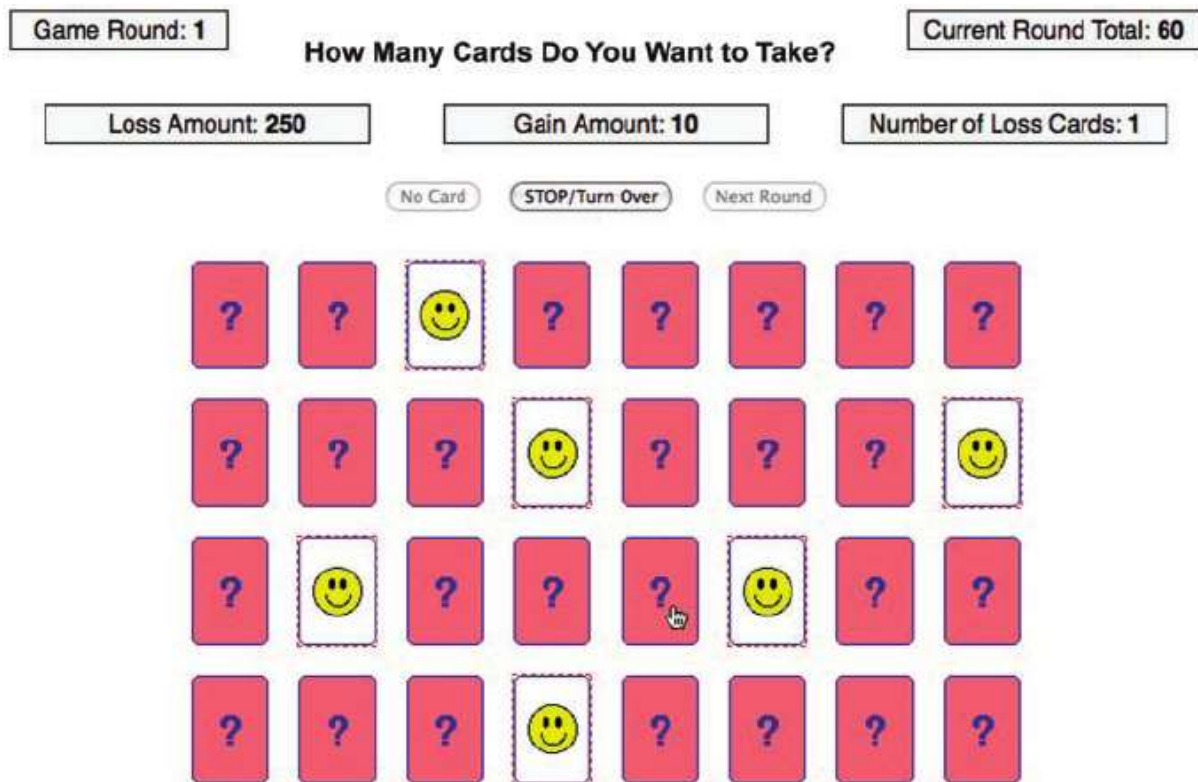


Figure 3 - Screenshot of the hot version of the Columbia Card Task (Figner et al., 2009).

The CCT has been utilised in several research domains and has found many applications. The CCT has found use in developmental psychology to examine age-related differences in risk-taking and highlight that adolescents tend to take more risks in the hot version due to heightened emotional reactivity (Figner et al., 2009). The CCT has also been used in clinical psychology to assess risk-taking tendencies in populations with affective disorders such as depression and anxiety where altered emotional processing may influence decision-making (Buelow & Blaine, 2015). Finally, the CCT has also been used in Neuroscience to investigate the neural correlates of risk-taking and differentiating between affective and cognitive contributions to risk-taking and decision-making processes (van Duijvenvoorde et al., 2015).

Despite the CCT's utility, the model has faced several criticisms. The first of many is its ecological validity. The artificial nature of the task may not accurately reflect real world risk-taking behaviours which limits the generalisability of the findings from this model (Pleskac et al., 2008). Another problem is the complexity of implementation. The dual versions (hot and cold) can yield wildly different results which can complicate the interpretation of what specific aspects of risk-taking are being measured (Figner et al., 2009). Like other models, this model falls short in considering individual differences such as personal traits, cognitive abilities and personality (Buelow & Blaine, 2015).

To improve the CCT, realistic scenarios could be incorporated, embedding the task within more ecologically valid contexts which would improve its applicability to decision making in real life. The task structure of the experiment could also be simplified which would reduce

complexity and would aid in clearer interpretation on results. Finally, the task could control for individual differences which would include implementing assessments of personality and cognitive function which could help account for variability in task performance. The CCT's distinction between affective and deliberate risk-taking offers effective and valuable insight when creating this project. Understanding how emotional and cognitive factors play into decision making can help create a risk-assessment tool that doesn't fall victim to the flaws of previous models

2.3 - The Role of Gamification

Gamification is the act of incorporating game design elements into non-game-based contexts to enhance user engagement and motivation. In psychological assessments, gamification aims to make tasks more engaging for users which improves the reliability of the data collected (Landers, 2014).

Traditional risk-taking assessments such as the BART and IGT can be perceived as monotonous which could potentially affect participant engagement and data validity. This issue is addressed with the implementation of gamification which aims to introduce elements such as immediate rewards, feedback and narrative contexts which can enhance user motivation and even simulate real-life decision-making scenarios (Hamari, Koivisto, & Sarsa, 2014). Introducing realistic situations can make a more immersive environment for the individual and make it easier for them to relate to the situation which would drive up user engagement.

However, with the implementation of gamification, there are many problems that arise with this too. The first is the "Over justification effect" which claims that there's a risk that external rewards in gamified assessments might undermine intrinsic motivation (Deci, Koestner, & Ryan, 1999). The next problem is validation concerns. It is crucial to ensure the gamified assessments accurately measure the intended psychological constructs. Game elements that are designed poorly may introduce biases or distract from the objectives of the assessment (Landers, 2014).

Incorporating gamification into this risk-assessment tool may fix some of the issues with previous risk-assessment models and help increase engagement and data quality. By incorporating gamification in a way that is appropriate, it can help create a tool that is both effective and enjoyable for users.

2.4 - Measuring Both Quantitative and Qualitative Data

In psychology, the introduction of integrating both qualitative and quantitative data (known as mixed methods) has become increasingly prevalent. The mixed methods approach combines the statistical robustness of quantitative data and combines it with the contextual strength of qualitative data. This provides a more comprehensive understanding of psychological phenomena (Creswell & Plano Clark, 2011).

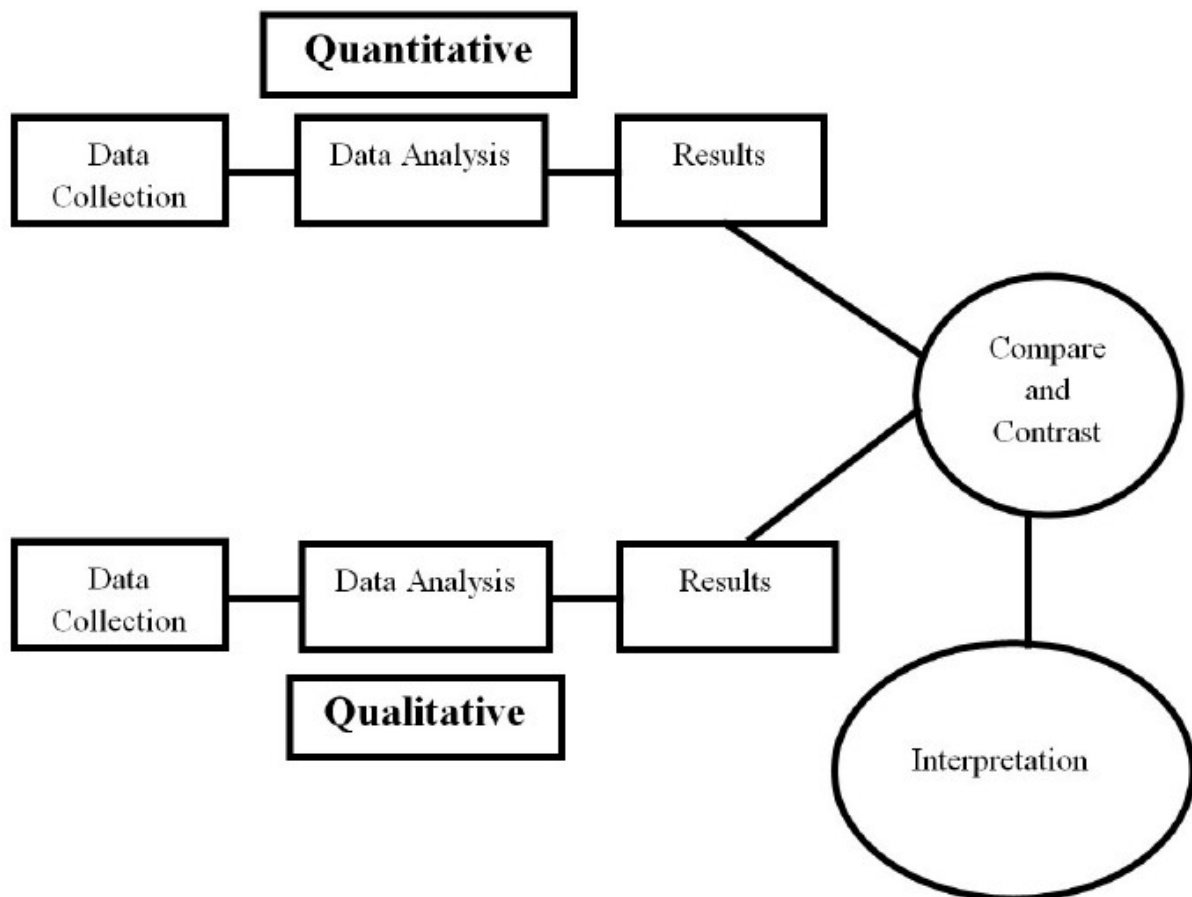


Figure 4 - A diagram of the mixed methods design approach

Quantitative methods such as standardised tests, structured questionnaires and experiment data allow for the measurement of variables such as reaction times, error rates and decision-making patterns in individuals. These methods assist in statistical analysis and comparisons across populations. However, the main disadvantage of quantitative data is that it may not fully capture the underlying reasoning and motivation behind these decisions.

Adversely, qualitative methods such as interviews, focus groups and open-ended survey questions, provide rich and descriptive data that fully outline the reasons behind the motivation for participants' choices and behaviours. In risk-taking assessments, qualitative data can reveal and provide insight into individual perceptions of risk, personal experiences and emotional responses that quantitative data might not fully cover (Keetharuth et al., 2018).

By employing a mixed methods approach to this task, researchers can triangulate data, enhancing the validity of findings and offering a more nuanced perspective on the reasonings behind risk-taking behaviour. The mixed methods approach also offers enhanced validity, the triangulation of data sources can corroborate findings which increases the credibility of results (Teddlie & Tashakkori, 2010). The mixed-methods approach is also extremely flexible, allowing for researched to explore unexpected findings and adapt their approaches accordingly

However, the mixed methods approach also comes with a lot of challenges which will need to be addressed with this project. The first is the complexity of the data. Designing and implementing mixed methods research requires careful planning to ensure there is a coherent link between the qualitative and quantitative data that is collected. Likewise, the mixed methods approach is also very resource intensive, collecting both types of data can be both time-consuming and complex which may require additional resources and expertise. Finally, there may be integration difficulties. Merging quantitative and qualitative data in a meaningful way can prove to be challenging, necessitating clear strategies for data integration (Fetters, Curry, & Creswell, 2013).

When developing a gamified risk assessment tool, incorporating both quantitative and qualitative measures is crucial to understand the underlying motivations and reasoning behind why decisions are made. This dual approach ensures a comprehensive evaluation of the tool's effectiveness and user engagement.

2.5 - Gaps in Literature and Justification for This Project

Despite extensive research in psychological risk assessments, several notable gaps exist in the study, a few of which have been discussed and identified previously. These gaps in research act as justification for this project, serving as things that this risk assessment tool will aim to improve and address.

The first of these gaps is ecological validity. Traditional assessments like that IGT and BART often lack the ability to relate to real-world scenarios and replicate real-world decision-making scenarios (Bechara et al., 1994; Lejuez et al., 2002). Standard assessments may also lack user engagement which could affect the accuracy of results (Landers, 2014). These assessments also fail to integrate gamification which has shown promise in enhancing engagement, its integration in psychological experiments remains limited (Hamari, Koivisto, & Sarsa, 2014). There is also a scarcity of assessments that integrate the mixed-methods approach which could help gauge a more holistic understanding of risk-taking behaviour in individuals. By building on all these previously discussed gaps in the field of risk-taking assessments, results can be made more accurate and be used over a wider range of fields and studies across multiple age groups and larger populations.

Chapter 3 – Design

3.1 – Overview of design goals

This project aimed to outline and design an interactive game which encourages natural risk-taking behaviour of individuals in a controlled and turn-based environment. The design principles applied in this game were guided by psychological theory and usability principles to fully ensure that data collection was valid and the experience remained intuitive while keeping the results appropriate and applicable for use by experimenters.

To mirror constructs that have been explored in existing studies such as the BART and IGT, the game presents the user with a range of choices which vary in riskiness. Each of the moves vary in accuracy and payoff which allows the user control of their level of risk based on the situation. The game also presents the user with “Elemental advantages” which add a level of uncertainty to the moves they make, allowing users to increase the level of risk they want to participate in. The game also aims to test how individuals make decisions under stress, implementing a mechanic to test how players interact with “near-losses” (Low-health situations.). This encourages behavioural expression of risk which aligns with the core idea of prospect theory (Kahneman & Tversky, 1979).

From the perspective of game design, the structure of the game promotes the use meaningful decision-making which is a core principle of engagement in game design (Schell, 2008). Additionally, care was taken to ensure that the game’s user interface aligns closely with usability heuristics (Nielsen, 1995) and follows valuable sensitive design principles (Friedman & Hendry, 2019) to ensure ethical transparency, particularly due to the nature of the project being a research tool.

3.2 – Game structure

3.2.1 – Overall system architecture

The game has been mainly developed on a JavaScript library called “p5.js”. p5.js allows users to create interactive environments that can be embedded onto web pages. The game has been embedded onto a regular webpage which has been designed with HTML and CSS. P5.js was chosen to create this game because it would be easier for users to interact easily with a

game if it is hosted on a website. This prevents users needing to download any programs or dependencies and allows them to quickly access the game without any troubles.

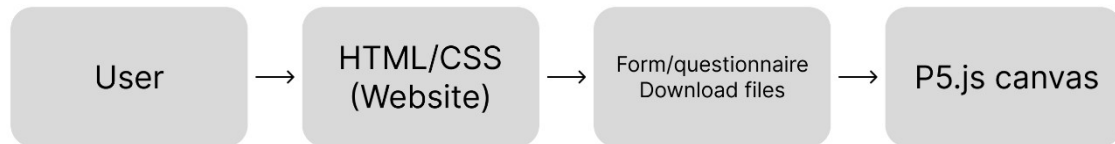


Figure 5- A system diagram outlining the flow of interactions

The HTML of the website contains all the logic for the website, this includes the links for any buttons, the structure for the website and any text and headers that need to be displayed to the user. In the case of this risk-taking experiment, the website also contains the p5.js canvas which is where the game will be played, the user's ID will also be displayed using HTML on the bottom of the webpage under the canvas. The website also displays the name of the game as well as its purpose, to the user.

The CSS of the website is used to style all the elements on the websites. This includes the use of colours, fonts, text-spacing and font-sizes. The CSS allows the website look accessible and

presentable for the user by making the buttons and text big enough to read and interact with, as well as providing a consistent motif and theme for everything.



Figure 6 - UI design of the intended look of the website with HTML and CSS

A UI design has been detailed in figure 6 which has been designed as a blueprint for the direction of what the website should ideally look like when the HTML and CSS has been implemented correctly. The p5.js canvas and game logic sit in the centre of the website with the interactable buttons being present above the canvas

The p5.js canvas focuses on rendering the game and displays it to the user. The JavaScript code is also contained inside the canvas, with the canvas acting as a visual representation of the JavaScript code. Inside the canvas, the JavaScript code is run which contains all the game-logic and necessary calculations for gameplay. The JavaScript also contains the logic for generating the User's ID when the page is loaded which is parsed onto the HTML and displayed on the web page. Additionally, the buttons that are displayed on the game are created using a mixture of HTML and JavaScript but are still contained independently within the p5.js canvas

Finally, the gameplay that is contained on the canvas is fully recorded for researchers to use and at the end of the game, a copy of the csv file is generated on the server on in the /sessions folder for the researchers to access. The server-side capabilities of the game are handled by

node.js, node.js allows the game to write csv files to the server when the game is completed. This file contains all the necessary gameplay data that the researchers may need to read through and use.

3.2.2 – Technology stack

- P5.js – A JavaScript library that was utilised in building the turn-based game system for the website. It enables the rendering of a canvas element within the HTML of the website which allows for user interaction via buttons and animations.
- HTML & CSS – Used to create and style the web interface which surrounds the p5.js canvas. HTML structures the page and defines the text and basic elements that are on the page while CSS ensures that the website is consistently themed and allows for accessibility and responsive design.
- Node.js (express.js) – Provides the game with server-side functionality for saving gameplay logs as .csv files to a /sessions folder which is located in the directory for the website server. The writing of files to a folder on the server allows for researchers to easily access files and retrieve data for analysis. Without node.js, files would not be able to be stored on the server as this is disabled by default by JavaScript. Researchers would not be able to access content as easily
- JavaScript (vanilla) – Used for the core logic handling on the website, including UI interactions, dynamic updates and interacting with both p5.js elements as well as HTML elements.

3.3 – User interface design

The user interface (UI) for this project was designed with the primary goal of accessibility, intuitiveness and minimal distraction from the core gameplay and game-loop. Given the context of the game being a psychological tool for studying decision making and risk-taking, UI decisions were made to balance clarity for experimental validity as well as engagement, to ensure user immersion and valid behaviour responses. A complex and overwhelming user interface could potentially lead to lowered understanding of the game's mechanics as well as a lack of focus on the main principles of the game which could lead to inaccurate and unreliable results.



Figure 7 - UI design mock-up for the game's main page

Figure 7 displays a mock-up user interface design for the game's main battle screen. This screen is where the main gameplay occurs and how the user the user will take risks and interact with the enemy. A priority has been put into minimalism and simplicity, allowing elements on the screen to be represented with abstract shapes and colours in order to prevent confusion and allow objects to be interpreted by a wide variety of individuals. The layout also reduces unnecessary UI clutter to prevent cognitive overload which is especially apparent in studies that concern decision-making. This principle also aligns with Nielsen's usability heuristic of "aesthetic and minimalist design" which warns that dialogues should not contain information which is irrelevant or rarely needed (Nielsen, 1995). The buttons are distinctively identified at the start of the game, with the three main tiers being presented to the player (Attack, defend, heal). These buttons are the biggest on the screen as these are the buttons that the player will press first before choosing their final move. The sub-buttons for the attack are presented near the main buttons but shown as smaller to indicate that they are a different kind of decision. This layout also adheres to core web usability principles that advocate for the "visual hierarchy and functional grouping" which ensures that players intuitively know where to focus their attention and how to act next (Norman, 2013).

The UI was also deliberately designed with large, high-contrast text and buttons to ensure visibility for users for visual impairments. These decisions have been made in line with W3C's accessibility guidelines (WAI, 2023) as well as game accessibility frameworks such as those that have been discussed in Bierre et al. (2005), who emphasise that many games can

unintentionally exclude users with disabilities through poor design choices. Due to my game aiming to be playable by a large variety of age groups, it is essential that I focus on making the game as playable as possible for the elderly and impaired

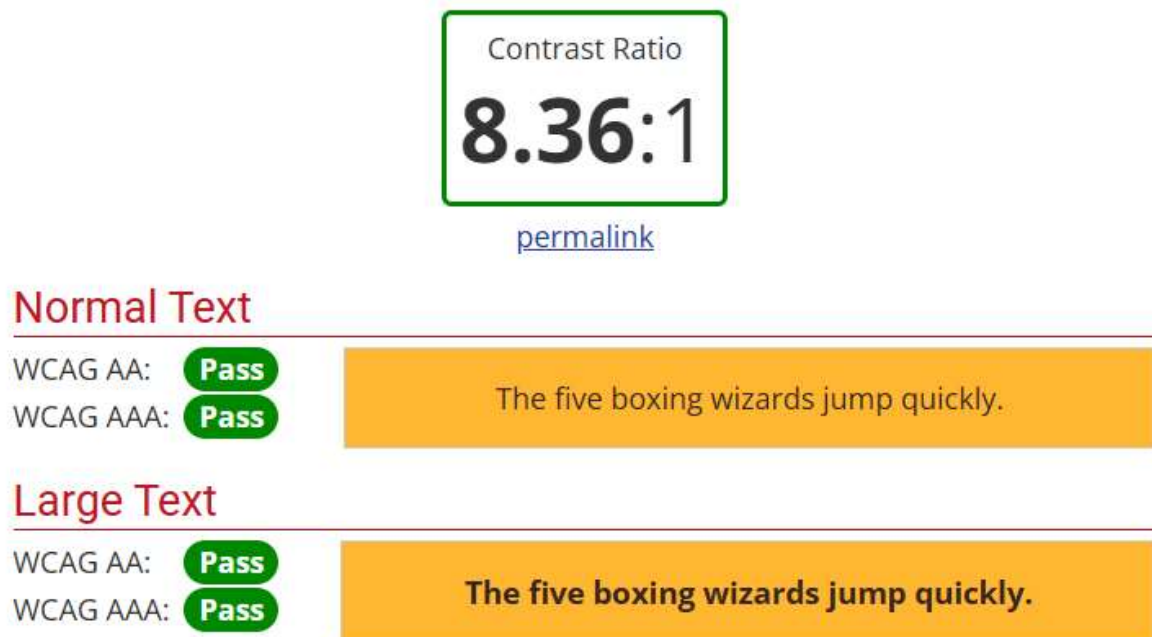


Figure 8 - Contrast ratio of text behind highlighted background, showing WCAG AA adherence

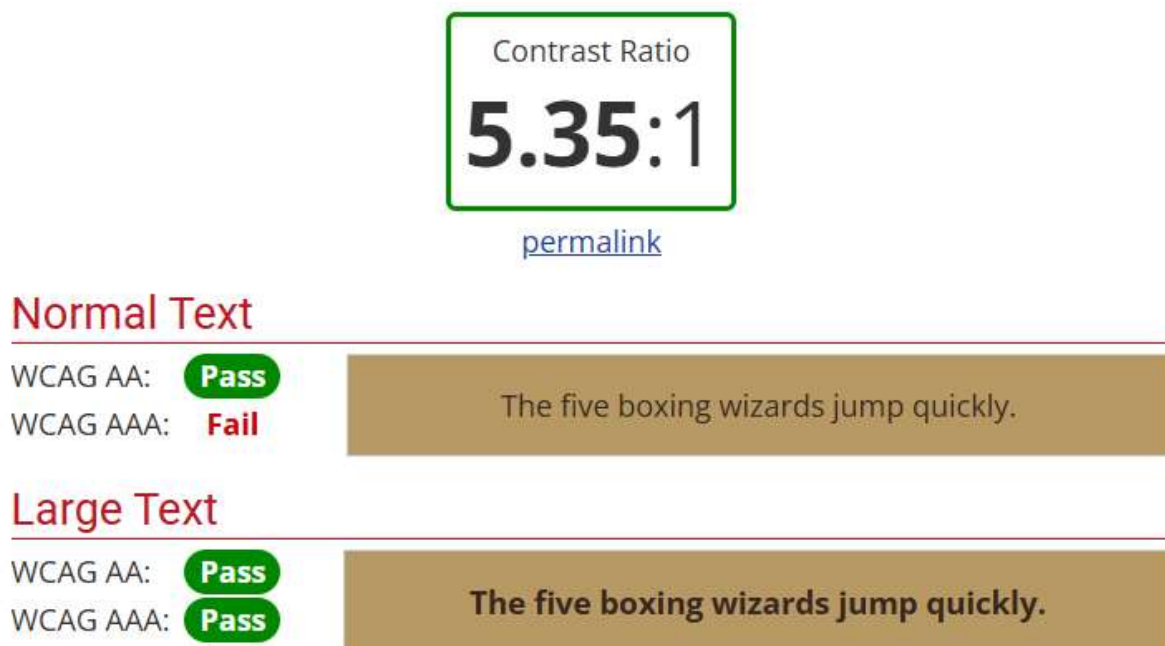


Figure 9 - Contrast ratio of text behind non-highlighted background, showing WCAG AA adherence

The buttons' contrast has adherence to WCAG AA accessibility principles. This allows a fair trade-off between accessibility and good visual design, letting impaired users read text easily and also allowing text to be coloured appropriately to fit within the theme and colour-motif of

the game. As well as this, buttons were also padded generously, allowing for users with motor impairments touch buttons easily and making the game easier to play for touchscreen input.

The game also provides immediate visual feedback upon player interactions. This includes button colour-changes, real-time updates to health bars and moves and floating numbers when damage or healing is done. These design elements are backed by usability literature, which highlights that responsive feedback helps people “feel in control” and builds a much stronger sense of interaction in a user interface (Shneiderman et al., 2010). The UI design also supports progressive disclosure, gradually revealing new elements to the user such as the final game state (win or lose) and the move that has been made only when needed. This reduces distraction and prevents premature data loss or confusion.

To align with ethical transparency principles (Friedman & Hendry, 2019), the user interface display the user’s unique player ID as well as the stated purpose of the game (A risk-taking experiment). This reinforces that the experience is part of an experimental context, reducing deception and supporting informed participation while also preserving ecological validity. The user interface also takes a non-diegetic approach to design, which separates interface elements from the game world. This has been done to ensure that experimental stimuli (move choices and health bars) are clearly presented to the player and understood properly. This is consistent with research by Iacovides et al. (2015) who found that non-diegetic interfaces can improve clarity and usability in mechanic-focused games.

3.4 – Risk taking mechanic justification

3.4.1 – Game loop and rules

At the start of the game, the player is presented with three main and distinct decisions, attack, defend and heal. Each move has 3 sub-moves which range in level of risk. Every turn the player and the Enemy AI both choose a move and when both moves have been chosen, they are revealed to each other and resolved respectively. It is important to note that even though the game is turn based, neither players move first, instead, moves are chosen and revealed simultaneously to both the player and the enemy and are resolved as such.

Attack moves allow the player to decrease the enemy’s health which is the main aim of the game. Attack moves are how the player progresses towards their goal of winning. There are three attack moves, each varying in the level of risk that they propose. The low-risk moves provides the player with low-damage but a guarantee at hitting the enemy and dealing damage. As the risk increases, the accuracy of the moves decrease which decreases the chance of the player hitting the enemy and increases the chance of the player’s move resulting in a “miss” which would have no effect on the enemy. Attack moves have a unique mechanic attached to them which is the option to attach an element to your move

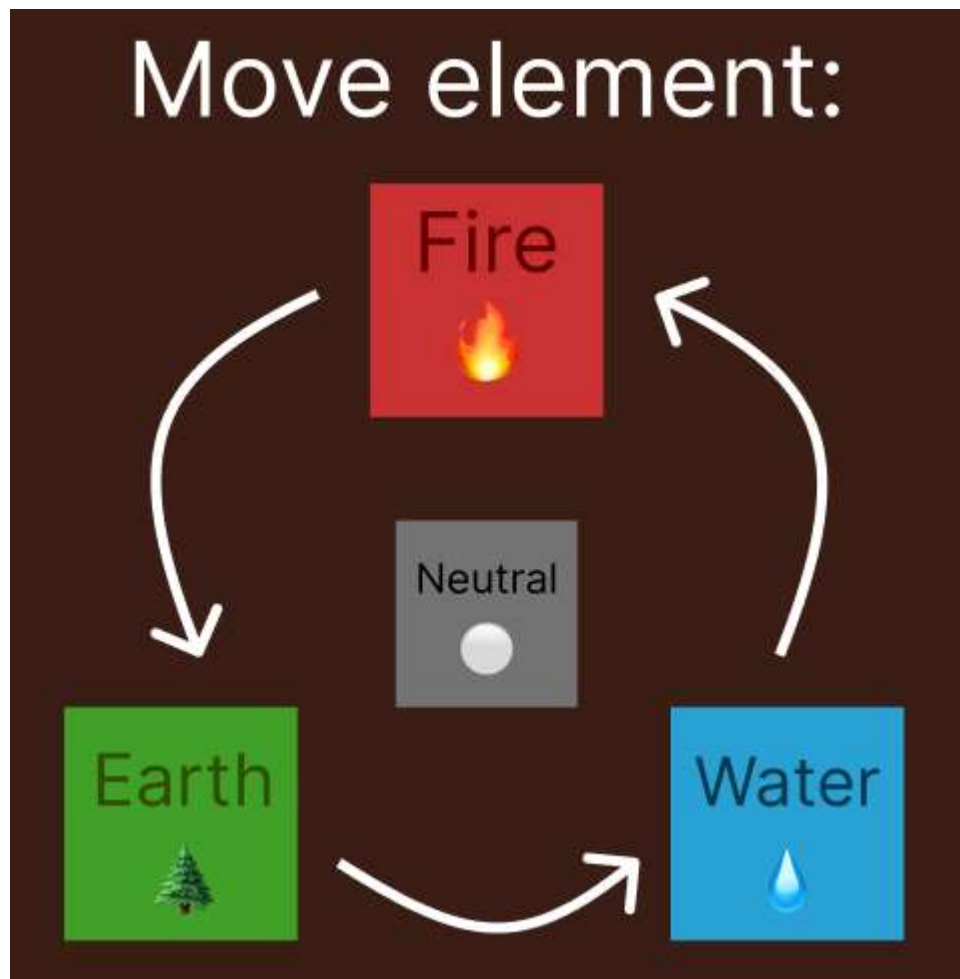


Figure 10 - A UI design of the element wheel

An attack move can be one of four elements: fire, water, earth and neutral. As outlined in figure 10, fire is strong to earth, earth is strong to water and water is strong to fire. If both users attack, the element that is strong against the other will do double the damage that it would usually do. Similarly, if your move is weak to another move, your attack would do half the damage which it would usually do. If the user wishes not to gamble their move, they can choose the neutral element which will allow their move to do the normal amount of damage that it would usually do, with no chance of doubling or halving the damage done. If the element of both moves is the same, the normal amount of damage will be done for both players.

Defend moves allow the player to block incoming attacks from the enemy by enemy for one move, decreasing the damage of the account by a percentage. Low-risk defence moves provide lower defence but provide a higher accuracy while the highest-risk move provides a high level of defence with a lower chance of casting successfully.

Healing moves allow the player to increase their health if damage has been taken. Like other moves, low-risk healing moves provide less healing and more accuracy while high-risk moves provide more healing but with less accuracy. Additionally healing moves have a

unique mechanic attached to them which decreases the accuracy of the next move based on the level of healing that the player chooses. Low-risk moves result in a lower accuracy penalty while high-risk moves result in a higher accuracy penalty. This risk penalty is only applied to the player if they successfully heal themselves.

Exclusive to the player, when the user drops below 15% health a “Berserk move” will be made present to the player which presents them with the option to use a move with extremely high damage and low accuracy but with a major punishment if the moves fails to hit the enemy. This move is entirely optional and can be ignored if the player wishes

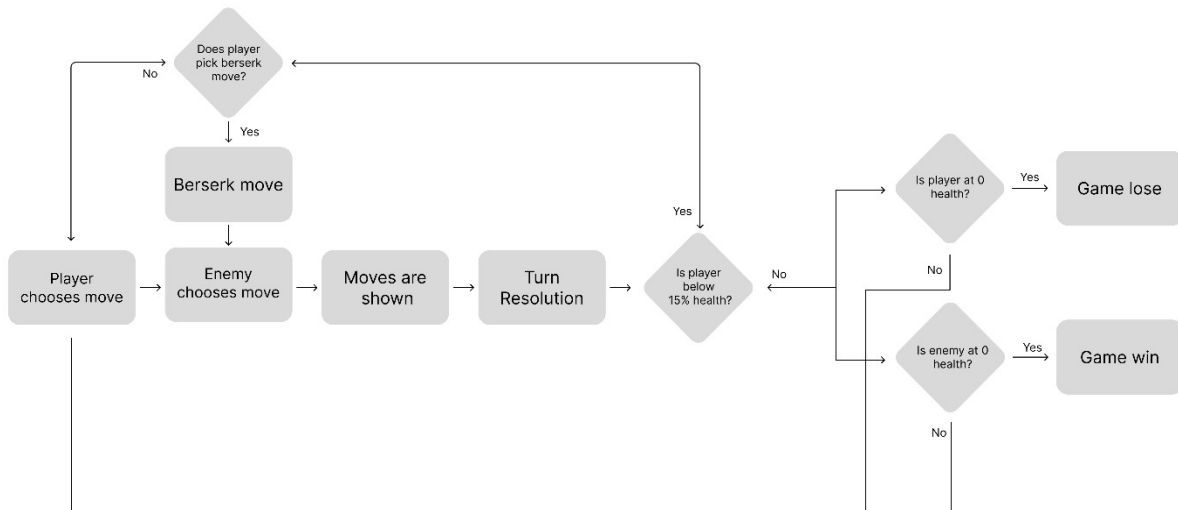


Figure 11 - A diagram outlining the gameplay loop

Figure 11 shows the intended gameplay loop for the experiment and emphasizes the importance of simultaneous choices and move resolution to ensure that everything is fair between the player and the AI.

The enemy opponent in the game is controlled by a custom AI system that is designed to simulate human-like strategic and reactive behaviour. The AI is designed to be dynamic and respond to the player’s decisions intuitively. This AI does not “cheat” by reading the player’s moves and reacts to information that another human playing would know, it uses information from past moves to make probabilistic decisions, mimicking how a human might infer and adapt in repeated rounds. Each turn, the AI selects one of three moves, Attack, defend and heal, using the following layered logic.

Health based prioritisation: If the AI’s current health falls below 35% it heavily favours healing (70% likelihood).

Pattern recognition: If the player has recently used many low-risk moves (e.g. Three or more low risk moves in a row), the AI is more likely to defend to soak up the incoming stream of damage

Aggression scaling: If the player has low health (Below 25%) or has healed often in recent rounds, the AI increases its aggression and favours attacking the player.

Default behaviour: If none of these conditions are met, the AI will default to a mild preference for attacking (60%) over defending (40%).

For move tier selection, there is another context-aware approach that the AI will use in order to determine the level of risk that it wants to take, similar to the type of move it would choose. For attacks, the AI chooses player health to scale aggression, the lower the player's health is, the higher-risk moves the AI will tend to choose to attack the player. For defends, the AI will increase the risk of defence based on its health. Higher risk defence will be chosen if its health is less than 50%. For heals, the AI prefers stronger when its health is lower

The enemy also utilises “elemental prediction” which applies for attack moves. When the enemy chooses an attack, it will base the attack's element on the element of the last 10 moves the player chose. It uses this information to find an appropriate counter-element based on the player's most common element. For example, if the player chooses fire more often, the AI is more likely to prefer choosing water moves. This form of pattern prediction adds a psychological mimicry layer which forces the user to adapt and encouraging counterplay

3.4.2 – Alignment with risk-taking theory

The mechanics of this game have been designed to provoke and measure real expressions of risk-taking behaviour by balancing design principles alongside psychological theory. This section details how the core mechanics mirror existing psychological literature and improve on known gaps in psychological assessment tools.

Each core move in the game (attack, defend, heal) has been designed to reflect the different cognitive and strategic motivations that come with decision-making and risk-taking under uncertainty.

Attack moves represent short-term reward-seeking behaviours. Low-risk attacks offer minimal but consistent and reliable damage, while adversely, high-risk attacks carry the potential for larger instant rewards at the cost of accuracy. This aligns well with “expected utility theory” and “prospect theory”, which suggests that individuals often weigh potential gains against possible gains under uncertainty (Kahneman & Tversky, 1979). Players that choose repeated high-risk attack moves may exhibit traits of sensation-seeking or impulsivity, reflecting research that has been conducted by Zuckerman (2007) into the psychology of risk preference.

Defend moves represent controlled risk-adverse behaviour. These decisions are often made by players that are aiming to reduce potential harm rather than seek immediate gain. This aligns with literature on defensive decision-making (Bechara et al., 1994). The trade-off between low-risk, low protection option vs high risk, high protection options also mirrors the strategic thinking evaluated in tasks such as the IGT.

Heal moves introduce a delayed consequence. Healing successfully reduces the accuracy of the next turn's accuracy, emphasizing long-term planning at the expense of short-term safety. This introduces a temporal risk component to the game which mirrors real-world decisions where efforts for recovery can temporarily impair responsiveness. This system maps well

onto the dual systems theory, which differentiates between impulsive and deliberate systems in the human brain (Steinberg, 2010). Together, this triad of attack, defend and heal creates a dynamic behavioural profile of each player that plays the game which reflects emotional regulation and risk appetite.

The berserk move appears as an option to the player when the player's health drops below 15% which offers a high risk, high reward option which could completely change the tide of the battle, and turn a losing game around in the player's favour. Similarly, the move could completely fail and push the player further into losing the game and potentially even just causing a full loss when the move backfires. This mechanic reflects prospect theory's concept of loss aversion. Individuals are far more likely to take extreme risks when trying to avoid a loss (Kahneman & Tversky, 1979). Similar to "hot" versions of the CCT which engage affective decision making (Figner et al., 2009), the "berserk move" places players in an emotionally heightened state which encourages decision making driven by stress and desperation. The naming of the move "Berserk" sparks a feeling of aggression and instability in the player, allowing them to understand that the nature of the move is reckless but powerful, which would play into the theme of the desperation move being desperate in nature. The berserk move also simulates the "near-miss" effect where players will narrowly avoid failure and are thus more motivated to continue risky behaviour (Clark et al., 2009). By including this mechanic in the game, real time emotional shifts are captured which may not be possible in models like the BART.

The elemental moves in the game allows for another optional level of risk for the player. Player can attach. Players can attach an element (fire, water, earth, neutral) to their attack. These elements interact with each other in a strength-weakness loop where effective moves deal double damage and ineffective moves will deal half damage. Choosing neutral allows the player to opt-out of this additional risk and deal the normal amount of damage without any gamble. This mechanic introduces optional volatility which offers the player the chance to self-regulate their level of engagement with risk-taking. This additional risk mirrors "framing effects" in behavioural economics where individuals' risk preferences are influenced directly by how choices are presented to them (Knežević & Jovanović, 2021). This system incentivizes and encourages behavioural flexibility which encourages behavioural flexibility and also mimics real-world conditions where risk may not be compulsory but can be heavily context-dependent. This also improves on models like the IGT which may not be fully ecologically valid and might not replicate real-world scenarios. Furthermore, it allows for strategic uncertainty where the opponent's prediction of your move can effect its outcome which is an element that is missing from most static risk assessments such as the IGT or the BART.

This risk taking assessment also utilises a custom AI system which adapts to how the player interacts with the game as well as certain conditions that can be dependant on the player's health and the enemy's health. This supports experimental validity in a multitude of ways. The AI increases ecological validity, the AI mimics real-life human behaviour which adds a level of realism and unpredictability while also rewarding forward-thinking and strategic thinking by attempting to counter the AI. This feature overcomes the critique of existing tasks

like the BART and CCT which are highly abstract and linear (Reynolds et al., 2006; Pleskac et al., 2008). The AI also increases pattern recognition and adaptive prediction. By analysing previous elements that have been used and countering them effectively, the AI simulated real-world cognitive processing like “predictive coding” and “meta strategy” which requires participants not only to react but adapt. This type of adaptive probabilistic AI ensures that learning effects are minimised which are a common issue in repeated trial studies such as the BART. This is done through continuous AI variability (Wallsten et al., 2005).

The mechanics of this game allow for an adaptive and layered approach for measuring psychological risk. Each individual system of the game maps onto validated psychological models and addresses gaps in traditional risk assessment tools by improving realism, emotional engagement and player agency. The model also largely expands on the concern of the lack of ecological validity of previous models.

3.4.3 – Risk alignment breakdown

This section outlines how each individual mechanic in the game aligns with psychological risk typologies. By deconstructing player decisions into categorised forms of risk, the game enables detailed analysis of individual risk preferences. This in-depth breakdown also demonstrates how the game expands on traditional risk assessment tools like the BART and IGT by incorporating multiple forms of uncertainty and decision-making.

Risk-taking is an inherently multifaceted construct. According to Weber, Blais and Betz (2002), risks can possibly be domain specific which involves different cognitive and emotional processes across contexts. Key risk dimensions that are relevant to this game are highlighted as follows.

Calculated risk – Logically derived risks where probabilities and outcomes are considered

Emotional or Desperation risk – Choices that are driven by affective states such as frustration, anger or panic.

Avoidance risk – Decisions that are based on minimising loss rather than maximising gain.

Strategic uncertainty – Voluntary risk where probabilities depend on unknown variables such as the opponent’s actions.

Deferred risk – Risks that have delayed consequences and require planning and foresight

The understanding of these categories provides a strong framework for evaluating and understand player behaviour, not just in terms of outcomes but also in the type of risks that they are willing to make.

Each core mechanic in the game targets a specific psychological risk type:

Low-risk attack	Calculated risk	A safe and repeatable action with predictable return. This reflects conservative strategies
High-risk attack	Calculated/ Voluntary risk	Offers a greater reward with lower accuracy. This invites deliberate risk selection.
Defend	Avoidance risk	Sacrifices momentum for damage control. This reflects a fear of immediate loss
Heal (With accuracy penalty)	Deferred risk	This rewards long-term survival but requires planning for the next-turn disadvantage
Berserk	Emotional/ Desperation risk	This is activated under stress, it highlights affective decision making in crisis states
Elemental Typing	Strategic uncertainty	Players choose their risk level and weather to gamble or not
Adaptive AI	Social/comparative risk	Players face evolving threats which mimics real world decision-making

Figure 12 - A table which maps game mechanics to risk forms

This mapping allows each player's behaviour to be coded alongside multiple axes of risk. For example, a player who repeatedly uses high-risk attacks may exhibit impulsivity or attention seeking traits, while a player who favours heal or defend actions may demonstrate an aversion to volatility and preference for control.

Traditional tools such as the BART and IGT capture specific types of risk in isolation. For example, BART focuses primarily on mechanical risk escalation where participants can inflate a balloon until it could potentially pop. The probability of failure is fixed and hidden and the player's decision is iterative but also narrow (Lejuez et al., 2002). The IGT involves hidden probabilities and long-term payoff recognition, this measures reward sensitivity and delayed learning (Bechara et al., 1994). While both are effective in controlled settings, they fall short in three areas. The first is a lack of voluntary opt-in and opt-out risk mechanics, the absence of emotional state triggers (the desperation mechanic) and no adaptive social dynamics which mimic realistic competition. In contrast, this game embeds layered and interrelated risk systems that allow players to toggle between conservative, impulsive and strategic risk-taking behaviours.

This classification system will serve as a coding guide and blueprint for gameplay analytics which will be explored further in chapter 5. By categorizing each choice by risk type it becomes possible to identify individual risk profiles across sessions. It also becomes possible to correlate specific patterns with survey-based self-reports as well as examining how players adapt to strategies when emotional triggers are introduced. Such analysis can reveal not just whether a participant is risk-prone but also the manner in which they take risks, strategically, emotionally and impulsively. This enhances the quality of both quantitative and qualitative data that is collected.

3.4.4 – Balancing risk and control

An essential aspect of any psychology focused game is ensuring that risk-taking is not forced, overwhelming, predictable or one-dimensional in any aspect. In this game, balance is maintained between risk and control across all mechanics, which supports the ethical integrity, validity of data and sustained user engagement. A key principle in experimental psychology is that participants must have the freedom to engage or abstain from risky choices. This is maintained in several ways.

Elemental typing is entirely optional; players can always choose neutral if they wish. Berserk mode is not triggered automatically; players must choose this choice under pressure. Attack tiers allow players to toggle between high and low risk moves each turn as freely as they wish. This aligns with work by Weber et al. (2002) who argue that perceived control over risk significantly influences risk-taking behaviour. Lack of control can lead to non-authentic, stress-driven responses which compromise the psychological validity of the study.

The game also avoids harsh punishment for risk, favouring strategic consequences. For example, the accuracy penalty for healing adds risk but does not completely nullify the benefit. The berserk penalty also only occurs on failure, offering conditional loss rather than guaranteed punishment. Such design reflects flow theory (Csikszentmihalyi, 1990) which emphasises the importance of feedback that challenged but doesn't completely overwhelm the player. Risk must feel earned but also manageable

Each action also comes with a predictable reward curve. Low-risk attacks result in low variance, high reliability. High-risk attacks result in high variance, low reliability. Defend has no damage but mitigates opponent's risk. Heal results in long-term benefit, short-term risk. This reflects the "risk-reward gradient" discussed in adaptive behavioural modelling (Steinberg, 2010). It helps ensure that player actions are traceable to intentional strategies rather than random or panicked decisions.

The game also includes mechanisms that help players recover from emotional extremes. Players may heal after going Berserk, creating a feedback loop between emotional risk and deliberate recovery. The AI does not punish berserk moves in predictable ways, avoiding deterministic feedback loops that could potentially frustrate players and invalidate natural decision-making.

The AI opponent plays unpredictably but fairly. It reacts to player trends over time but cannot read current actions. This creates a dynamic challenge the escalates or de-escalates based on

the player's style. There is no artificial advantage which could otherwise invalidate comparative risk-taking metrics. This strikes a balance between consistency and adaptability, satisfying research demands for valid social simulation without falling into determinism (Wallsten et al., 2005).

3.5 – Data collection design

This chapter outlines the methods that were utilised in order to collect both gameplay and self-reported data to analyse risk-taking in participants. This method of data collection aligns with the mixed methods approach which was discussed in chapter 2.4, enabling both motivational analysis and behavioural insights into why individuals made certain decisions during gameplay

3.5.1 – Automated gameplay logging

During gameplay, the automated system records every key decision made by the player. These include the following pieces of information

- Type of move selected (Attack, Defend, Heal)
- Risk level of the move (Low, Medium, High)
- Element chosen (Fire, Water, Earth, Neutral, n/a)
- Whether berserk move was activated
- Outcomes of each move (Hit, miss, success)
- The amount of damage done to the enemy, if any
- The move number

These values are stored server-side in a .csv file format, with each row representing a single turn and each column representing a unique set of data. This allows for structured quantitative analysis of gameplay patterns and risk-taking tendencies.

3.5.2 – Post-game questionnaire design

To complement gameplay data, a structured post-game questionnaire will be administered through google forms. This form will capture both quantitative and qualitative data in alignment to the mixed methods approach outlined earlier. This form will capture data such as players' self-reported perception of decisions, risk tolerance, emotional reactions and justifications behind why decisions were made. The questionnaire will also collect optional and non-personal identifiable information about the player such as age, gender and experience with games to allow for subgroup analysis

Question	Scale
I generally prefer low-risk strategies	Strong disagree - agree
I took more risks in this game than I usually do	Strong disagree - agree
I felt emotionally affected when I was close to losing	Strong disagree - agree

I used the Berserk move because I felt that I had no other choice	Strong disagree - agree
I enjoyed the element typing mechanic	Strong disagree - agree
I avoided using elemental attacks due to the risk	Strong disagree - agree
I felt the enemy AI was fair and challenging	Strong disagree - agree
I felt in control of my decisions throughout the game	Strong disagree - agree

Figure 13 - A table showcasing all quantitative responses (Likert scales) in the questionnaire

The qualitative (open ended) questions collected are as follows:

- Why did you choose to use Berserk (or not)?
- How did you decide which element to use, if any?
- What was your general strategy in the game?
- Did your playstyle change when you were close to losing? Why?
- Do you have any feedback about the game or mechanics?
- Are there any other additional comments you'd like to add about your gameplay?

3.5.2 – Ethical and technical considerations

All data collected is fully anonymous and only identifiable through the participant ID. No personal or identifiable information is collected about the participants. Along with this, a clear consent statement is outlined in questionnaire which outlines data use. Questions are also fully voluntary, and the participants may skip over questions as they please. Data is stored securely on password-protected machines only accessible by the developer.

3.6 – Design iterations and improvements

Throughout development, the game went several iterations and revisions in response to testing, gameplay imbalances and self-review. This chapter outlines the most significant changes and the rationale behind them.

3.6.1 – Move system refinement

During initial development, when a player chose a move, the result of the move was resolved before the enemy chose a move. This caused the issue of the enemy's AI reacting after the player which often gave it an advantage over the player. For example, the AI could only benefit from the elemental attacks mechanic as the player's attack was registered before the enemy chose an element. To resolve this issue, the game's design was altered to use a simultaneous resolution system for moves instead of sequential

3.6.2 – Elemental mechanic redesign

When initially designing the elemental system, there were only three elements and not the option of the neutral element. This created a forced risk for players that they could not opt out of which could introduce some experimental invalidity. To fix this, the neutral element was added which allowed the player to freely opt out of the risk if they wished.

3.6.3 – UI and accessibility changes

In this initial design for the game, UI icons were small and unintuitive for the elderly and touchscreen devices. This created an accessibility problem for the less impaired. To combat this issue, the icons in the game had their size increased and the colours were chosen to have a good contrast to the background while also providing a clear and consistent theme. Colour coded risk levels were also added as a visual indicator for risks

3.6.4 – Enemy being too easy

During initial development, the enemy AI was far too easy, spamming low risk moves could easily have you win the game and the player's ability to win was not based on their ability to adapt and use counterplay against the AI. Subsequently the AI was designed to challenge the player and increase engagement for the player.

3.6.5 – Ability to change moves

At first, when the player chose a move, they had to fully go through with it, which didn't allow the player room to make comprehensive decisions. If the player changed their mind about a decision, it was not possible for them to change their move which could potentially discourage complex thinking. To combat that, a simple system was designed to let the player choose through moves in the game freely until they were ready to lock in their move

3.7 – Ethical design considerations

The development of this game as a psychological assessment tool required careful and thorough attention to ethical principles throughout its design, implementation, and deployment. The following section outlines how the project ensures participant safety, data protection, and equitable accessibility, in accordance with the British Psychological Society's ethical guidelines (BPS, 2021) and MMU's Research Ethics procedures (Approval ID: 76392).

3.7.1 – Informed consent and voluntary participation

Participants are presented with a short consent statement before the post-game questionnaire, informing them that all gameplay and survey data are fully anonymous and optional and used solely for academic research purposes. The use of a non-identifying participant ID ensures that the user can withdraw from the experiment at will without being identified or personally traced. No personally identifiable information is collected about the user unless they

personally provide it (Age, gender). This upholds the principle of autonomy, ensuring that users are free to participate without coercion or risk identification.

3.7.2 – Data anonymity and secure storage

Logs for the gameplay are stored in a .csv format using randomly generated session IDs from the p5.js JavaScript code. All this information is stored in a secure directory on a password protected device with limited access. All exported data is handled under MMU's governance protocols, and the design adhered to the GDPR principle of data minimisation, collecting only data that is directly relevant to research aims.

3.7.3 – Avoiding psychological distress

Given the game's nature of measuring of measuring risk-taking behaviour, this may include failure and loss which could potentially cause mental discomfort. Steps were taken to ensure to minimise psychological discomfort.

- No real money or punishment is involved in any decision
- Berserk attack is voluntary and reversible
- UI feedback is neutral in tone, no red “failure” screens or anything overly alarming is used
- The player can exit the game at any time without penalty or data being saved
- Game elements are abstract and do not identify heavily with personal experiences

The game avoids mechanics that can be deceptive, as well as scripted losses, ensuring that participants are never tricked or manipulated by the gameplay. This aligns with ethical standards against deception unless scientifically justified (APA, 2017)

3.7.4 – Inclusive and accessible design

The game is designed to be accessible as possible, reflecting MMU's EDI (Equality, diversity and inclusion) principles. They have been implemented as follows:

- Colourblind friendly UI: A red and green contrast was avoided for making key decisions and was only implemented for small indications
- Simple and readable fonts were utilised with labelled buttons to ensure cognitive accessibility
- Time-limited decisions were not implemented to reduce stress for neurodivergent players
- The game is accessible through browsers without any installation from the user

The game avoids assumptions about prior gaming knowledge which makes it approachable for a wide range of participants. The game has been made accessible for non-gamers by including a simple tutorial which details the rules and mechanics of the game

3.7.5 – Ethical oversight

Ethical approval for the project was granted under MMU reference #76392. Participants are invited to provide feedback via the Google Form, and any discomfort or questions can be raised directly with the researcher through a voluntary contact field at the end of the questionnaire.

Chapter 4 - Implementation

4.1 – System architecture

This game is built using HTML, CSS and p5.js for rendering visuals, animations and capturing the player's inputs. The core gameplay loop which includes turn resolution, move selection, logic for elements and berserk functionality is handled entirely within the browser. P5.js also creates the final .csv file which is sent to the server directory via node.js

- User interactions including buttons clicks, move choices and decisions are processing through the central p5.js sketch within the website
- Real time feedback is provided through visual UI changes, highlighted buttons, health bars and text prompts under the player and enemy.
- The game simulates simultaneous turns by taking input from both the enemy and player at the same time and only processing the turns when both have chosen their move. Comparisons are made in a comparative fashion

This separation and technical line-up allows for fast iteration and direct observation of behavioural patterns while also maintaining local responsiveness

The backend for the game has been implemented with Node.js with custom endpoints for receiving and storing the necessary gameplay data

- Each move made by the player is send via HTTP post to the backend of the game
- The necessary data is stored in structured .csv files (one per session) which enables quick parsing and analysis for Python, R or Excel.
- Data fields include the move type, element (if it's an attack move), risk level, move outcome and round number

The use of .csv over a traditional database system such as SQLite was chosen in order to prioritise simplicity, portability and alignment with typical analysis tools used in psychology (Lehmann et al., 2020). Each game session generates its own data file when the game concludes, this reduces complexity in tracking participants and supporting anonymised branch analysis

While the AI logic executes on the frontend, its behaviour is designed to be dynamic and pseudo adaptive which allows it to mimic human decision-making patterns. The AI does not access player inputs directly but uses past player behaviour to make predictive counter-choices. This structure provides fairness and integrity in this study while also simulating social unpredictability which is a necessary condition for an ecologically valid risk-taking analysis (Figner & Weber, 2011).

4.2 – Key modules and implementation highlights

The game's architecture consists of several modular systems which each fulfil a specific functional or psychological purpose. This section outline the implementation of core mechanics and systems that underpin gameplay, data capture and adaptive behaviour.

The core gameplay loop is governed by a simultaneous turn resolution loop which is implemented in p4.js. Each round the player selects a move and move type. The enemy then picks its move based on past player trends and then a `compareMoves()` function evaluates both actions before applying damage, healing or any defensive modifiers. This helps avoid the bias of traditional turn-based systems and more closely simulates real-time decision environments. The resolution system also handles edge cases such as simultaneous defeat or invalid input with controlled fallback states

```
function moveRegister() {
  //player choose their move by pressing buttons

  //enemy chooses their move
  enemyChooseMove();

  if (moveElement !== "n/a") {
    playerElementHistory.push(moveElement);
    if (playerElementHistory.length > 10) {
      playerElementHistory.shift(); // limit memory to last 10 rounds
    }
  }

  //the comparison to the moves happens here
  //calculates defense of both parties first
  defendLogic();
  elementLogic();

  //Player move is applied to the game
  attemptMovePlayer(moveType, move, moveElement);

  //Enemy move is applied to the game
  attemptMoveEnemy(enemyType, enemyMove, enemyElement);
}
```

Figure 14 - A snippet of the Move Register function that shows how player moves are compared at the same time and applied simultaneously

As shown in figure 15, both the enemy and player choose their move, and the game will compare that information before applying the final damage values to the enemy and player. Each action has three risk tiers (Low, medium and high) which affect accuracy, impact, and any penalties given to the player for that specific move. Moves are resolved through the `defendLogic()` and `elementLogic()` functions as defence and element moves must be specifically calculated fairly to apply damage in a consistent manner. The enemy's AI is contained entirely in the `enemyChooseMove()` function, and once the move is chosen, it is applied to the game with the `attemptMoveEnemy()` function. The AI chooses through a pre-determined set of moves based on a specific set of criteria which is outlined within the function

```

function enemyChooseMove() {
  const moveTiers = {
    attack: ["Low Attack", "Mid Attack", "High Attack"],
    defend: ["Low Defend", "Mid Defend", "High Defend"],
    heal: ["Low Heal", "Mid Heal", "High Heal"],
  };

  // % Health logic
  const enemyHealthPercent = enemy1.enemyHealth / enemy1.enemyMaxHealth;
  const playerHealthPercent = player1.playerHealth / player1.playerMaxHealth;

  // Analyze player's recent strategy
  const recentMoves = moveListPlayer.slice(-5);
  const playerLowRiskSpamming =
    recentMoves.filter((m) => m.includes("Low")).length >= 3;
  const playerHasHealedRecently =
    recentMoves.filter((m) => m.includes("Heal")).length >= 2;

  // AI decision-making logic
  let type = "attack";

  // Heal if health < 35%
  if (enemyHealthPercent < 0.35 && random() < 0.7) {
    type = "heal";
  }

  // Defend if player spams low damage
  else if (playerLowRiskSpamming && random() < 0.5) {
    type = "defend";
  }

  // Pressure player if they're healing often or low HP
  else if (

```

Figure 15 - A snippet of the EnemyChooseMove() function, showcasing the pre-chosen moves and some of the AI logic underneath

For instance, if the player frequently chooses Fire, the AI is more likely to select Water. This creates a strategic pressure similar to human anticipation, without compromising fairness. The AI operates asynchronously from the player's input, preserving ecological validity.

Each round's data is sent to the Node.js backend where a custom logging module formats and stores all the game data into an organised .csv file. The creation of the .csv file is handled with the p5.js frontend with the exportGameDataCSV() function. When this game file is created and the game ends, this file is parsed over to the node.js backend for proper error and file handling in-case the file fails to download or process properly

```

const express = require("express");
const fs = require("fs");
const path = require("path");
const app = express();
const PORT = 3000;

// Middleware to parse JSON
app.use(express.json());
app.use(express.static(__dirname)); // Serve static files (index.html, etc.)

// POST route to save CSV to /sessions
app.post("/save-csv", (req, res) => {
  const { filename, content } = req.body;

  const savePath = path.join(__dirname, "sessions", filename);
  fs.writeFile(savePath, content, (err) => {
    if (err) {
      console.error("Failed to save file:", err);
      return res.status(500).send("Error saving file.");
    }
    console.log(`Saved CSV as ${filename}`);
    res.sendStatus(200);
  });
});

// Start server
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});

```

Figure 17 - a snippet of the server.js file, showcasing the backend

```

function exportGameDataCSV() {
  let csv = "PlayerID,MoveNum,Role,MoveType,MoveTier,Element,Outcome,Value,Success\n";

  for (let i = 0; i < moveListPlayer.length; i++) {
    const [, moveType, move, element, outcome] = moveListPlayer[i].split(", ");
    const value = outcome.includes("damage") || outcome.includes("Healed") ? outcome.match(/\d+/)[0] : 0;
    const success = !outcome.includes("fail") && !outcome.includes("Miss");
    csv += `${playerID},${i + 1},Player,${moveType},${move},${element},${outcome},${value},${success}\n`;
  }

  for (let i = 0; i < moveListEnemy.length; i++) {
    const [, moveType, move, element, outcome] = moveListEnemy[i].split(", ");
    const value = outcome.includes("damage") || outcome.includes("Healed") ? outcome.match(/\d+/)[0] : 0;
    const success = !outcome.includes("fail") && !outcome.includes("Miss");
    csv += `${playerID},${i + 1},Enemy,${moveType},${move},${element},${outcome},${value},${success}\n`;
  }

  // NEW: Send to backend instead of downloading
  fetch('/save-csv', {
    method: 'POST',
    headers: {
      'Content-Type': 'application/json',
    },
    body: JSON.stringify({
      filename: `session_${playerID}.csv`,
      content: csv
    })
  })
  .then(response => {
    if (!response.ok) {
      throw new Error('Failed to save CSV on server.');

```

Figure 16 - exportGameDataCSV() function in the p5.js sketch

Figure 16 and 17 showcase how the csv file handling is conducted within the code of the

program. The CSV file is named according to the player's ID who generated the data to help keep the data consistent and easy to find and keep each game personally identifiable with whoever played it.

4.3 – Testing and debugging approaches

Testing and debugging were carried out in an iterative manner throughout development to ensure that the game behaved predictably, logged data in a reliable way and produced valid results for psychological analysis. The project relied on a combination of manual playtesting, console logging, edge case simulations and in-browser network inspection tools.

The main form of playtesting that were carried out was manual play testing which involved repeated full playthroughs by the developer. These tests focused on the following areas:

- The first was button logic, ensuring that the buttons displayed their intended output and completed their specific intended job. This included buttons for the different attack tiers and element selection
- Simultaneous move resolution was important, a series of games were played in order to ensure that move resolution occurred successfully and fairly, with both the player and enemy being able to benefit from attack multipliers.
- Game state transitions were tested to make sure all the unique screens were accessible for the player when the correct conditions are met

This manual testing helped expose flow inconsistencies such as problems with the berserk move triggering prematurely and edge cases where incorrect button states would persist between turns

Console logging was also implemented in order to test that things that might not be directly visible are working correctly. This includes the states of hidden variables (accuracy penalty, attack multipliers etc.) These console logs will print to the p5.js editor to be viewed.

`Console.log()` outputs were used for accuracy and penalty calculations, health values pre and post action, enemy AI decision reasoning and elemental damage multipliers.

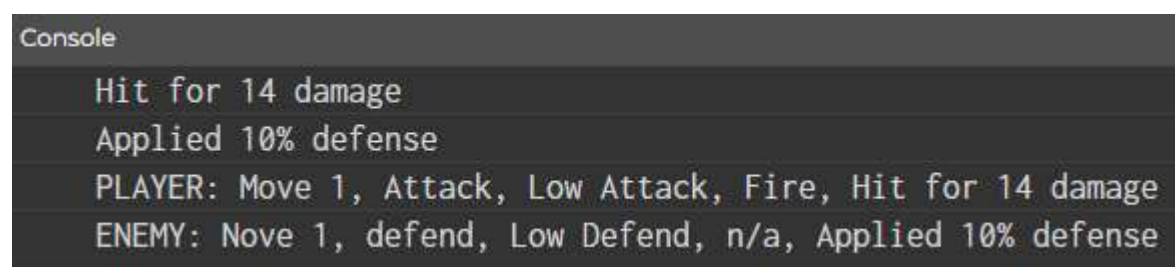


Figure 18 - The console log of the p5.js web editor showcasing the outcome for enemy and player moves, used before visual interface was added

In addition to this debug hotkeys were applied to a testing build o the game which would display information on request such as move logs, player ID and penalty values without breaking the gameplay loops. The nature of these logs being activated on a button pressed allowed the viewing of elements before and after they were changed to see how a function

would affect the value of a variable in the code. In the final version of the program, these were removed or commented out to make the code concise and clean and to prevent players from doing anything that they weren't intended to do

The backend server.js file was tested using postman to simulate API requests to the /save-csv endpoint. Along with this, manual tampering tests were conducted by sending malformed JSON or empty payloads to verify error handling. Finally, file system checks were put in place, confirming that .csv files were stored under the correct session name and contained well structured rows. Each exported .csv file was opened in both excel and a text editor to validate file formatting and ensure field consistency for later data analysis

	A	B	C	D	E	F	G	H	I
1	PlayerID	MoveNum	Role	MoveType	MoveTier	Element	Outcome	Value	Success
2	46YSAI0M95	1	Player	Attack	High Attack	Fire	Hit for 45 damage	45	TRUE
3	46YSAI0M95	2	Player	Attack	High Attack	Fire	Missed	0	FALSE
4	46YSAI0M95	3	Player	Attack	High Attack	Fire	Hit for 41 damage	41	TRUE
5	46YSAI0M95	4	Player	Attack	High Attack	Fire	Missed	0	FALSE
6	46YSAI0M95	5	Player	Attack	High Attack	Fire	Hit for 41 damage	41	TRUE
7	46YSAI0M95	6	Player	Attack	High Attack	Fire	Hit for 36 damage	36	TRUE
8	46YSAI0M95	7	Player	Attack	High Attack	Fire	Hit for 45 damage	45	TRUE
9	46YSAI0M95	1	Enemy	attack	Low Attack	Neutral	Enemy hit for 20 damage	20	TRUE
10	46YSAI0M95	2	Enemy	defend	High Defend	n/a	Defense failed!	0	FALSE
11	46YSAI0M95	3	Enemy	defend	Low Defend	n/a	Applied 10% defense	0	TRUE
12	46YSAI0M95	4	Enemy	defend	Mid Defend	n/a	Applied 20% defense	0	TRUE
13	46YSAI0M95	5	Enemy	defend	Low Defend	n/a	Applied 10% defense	0	TRUE
14	46YSAI0M95	6	Enemy	defend	Mid Defend	n/a	Applied 20% defense	0	TRUE
15	46YSAI0M95	7	Enemy	heal	High Heal	n/a	Heal failed	0	FALSE

Figure 19 - An example .csv dataset produced from gameplay

Despite all of this, some design limitations were noted and managed accordingly. Occasional CSS misalignment on smaller screens was mitigated by fixed canvas positioning and centre alignment. Duplicate submissions were prevented using a hasExported flag in the p5.js code before csv export. AI behaviour while not deterministic, repeated patterns. This was improved by adding probabilistic variation based on recent player trends

4.4 – Deployment and accessibility

The project is structured as a self-contained folder with the following components:

- An index.html file and a style.css file to handle layout, UI and theming of the website surrounding the game
- A sketch.js file which contains the entire code for the frontend logic of the game
- A server.js file which contains node.js backend API for saving .csv files
- A google forms link which is embedded into index.html for a seamless transition to the post-game questionnaire.

Users launch the game by opening the index.html file in a browser through the node.js server. When the round is completed, data is sent via POST to the local server running the server.js file. All responses are logged into .csv files which are ready for batch analysis. The game was

also designed to be lightweight, visually clear and cognitively inclusive. This means that the game is fully functional on desktop browsers, and no installation is required to run the game. The users only need internet access and a keyboard/mouse to play the game. Colour contrast and hover feedback were tuned using CSS classes for both the p5.js sketch and the surrounding HTML file. Elements were labelled and spaced fairly to reduce cognitive strain and improve clarity for the user. To improve experience for all participants, no time limits were enforced during play. The UI feedback is gentle and avoids overwhelming visual effects, supporting inclusion for users with sensory sensitivities or slower processing speeds.

Chapter 5 - Analysis

5.1 – Overview of data collected

Data was collected through two primary methods. The first is in-game logging and the second is a post-game questionnaire. Each unique gameplay session generated a structured .csv file recording the player's decisions per round, including move types, risk tiers, elements used, health values, and Berserk move usage. This was paired with a Google Form that gathered quantitative ratings (e.g. perceived risk, emotional response) and qualitative feedback on player strategy and decision-making. This dual approach enables both behavioural measurement and self-reported insight, aligning with the project's mixed-methods framework and allowing triangulation of player risk profiles (Fetters et al., 2013).

5.2 – Quantitative analysis plan

Quantitative data analysis will focus on the feedback given by players by the post-game questionnaire and this data will be cross-referenced with the data generated by the .csv file when the game concludes. The self-evaluated data through the questionnaire can be compared to the data from the .csv to determine whether the player understood the level of risk they were taking and their own understanding of perceived risk. In terms of the .csv file, the data can be used to gather an understanding of the player's general level of risk as well as their play style in response to the enemy AI.

5.3 – Qualitative analysis plan

Qualitative data was collected through open-ended questions in the post-game questionnaire. These questions invited players to reflect on their choices, emotions and overall strategies including information like why they did or didn't use the berserk move, how they selected elements and avoided them, whether their strategy changed when close to losing and general feedback about the game's mechanics.

Responses will be analysed using thematic analysis (Braun & Clarke, 2006) which involves reading through all responses to familiarise with the data, identifying recurring concepts or reasoning patterns (E.g. Panic, calculated risk, avoidance etc.) and grouping these into broader themes such as desperation-driven decisions, risk aversion or strategic adaption. These themes will be cross-referenced against the gameplay logs. This allows depth to be

added to the behavioural interpretation and reveals the psychological motivations behind actions that may appear inconsistent at the surface level

The analysis will be manually read through for a small sample (10 responses) due to the limited scale of the study; this allows for a manageable yet insightful exploration of player reasoning

5.4 – Triangulation of expected patterns

The final stage of analysis involves triangulating quantitative gameplay data with qualitative self-reports. By comparing what players did with what they said, the study can assess the alignment between observed behaviour and self-perception which is a key concern in behavioural psychology (Smith & Firth, 2011). For example, players who self-identify as risk-averse but consistently select high-risk attacks may reflect either misjudged self-awareness or emotionally driven adaptation during play. Similarly, frequent element use could correlate with self-reported strategic experimentation. This synthesis allows for a more holistic understanding of risk-taking by recognising both the mechanical outcomes and psychological motivations behind them. While the scope is exploratory, the triangulated design enhances the validity and interpretability of the findings and supports future extensions of the tool in larger studies.

5.5 – Pilot testing and observed feedback

To assess the effectiveness and usability of the game, 10 volunteers were recruited for informal pilot testing. These participants included, friends and family members, with varied levels of gaming experience. Each tester played at least one full game session and provided post-game reflections through informal discussion and the post-game questionnaire.

5.5.2 - Observed Quantitative Gameplay Behaviour

Gameplay logs and live observation revealed several consistent patterns. The first of which is that participants began the game with low and medium risk attacks and only started using higher risk attacks once they became more familiar with the game flow. Berserk mode was often avoided unless the player had “mentally accepted” that they were close to moving, or that they had no other option to help them win. The neutral element was the most used element, suggesting that participants preferred predictability over strategic uncertainty. One participant began using elemental counters in the fifth round only after realising their impact and benefit which could highlight a learning curve in understanding risk-rewards

Do you have past-experience with games?

10 responses

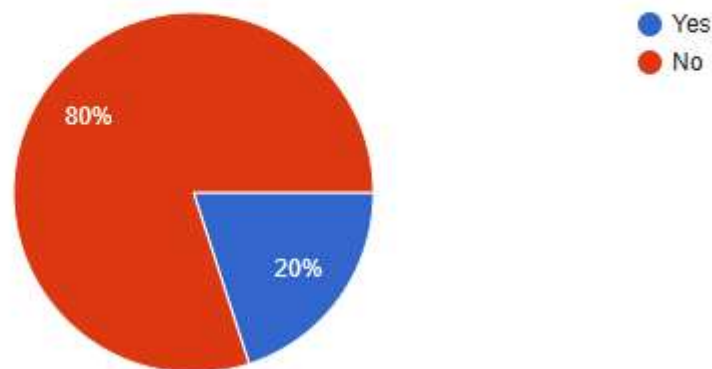


Figure 20 - Google form results showing participants' experience with games

I felt emotionally affected when I was close to losing

[Copy chart](#)

10 responses

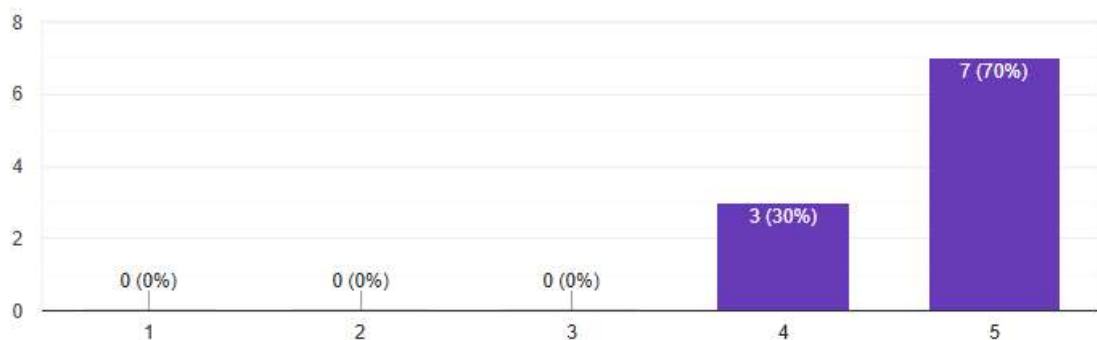


Figure 21 - Google form results showing if participants are emotionally affected

As shown in figure 20 and 21, most participants had no experience with games, despite this, most participants felt emotionally affected when they were close to losing. This could potentially indicate the game's engaging nature and ability to hook in participants regardless of their background. Similarly, no players reported discomfort or psychological distress, so this data suggests that the emotional responses were positive and aided individuals in making decisions

I felt in control of my decisions throughout the game

 Copy chart

10 responses

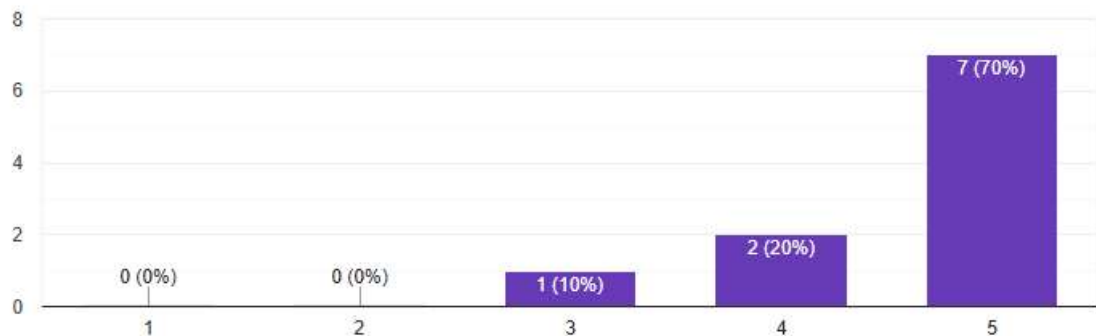


Figure 22 - Google form results showing if participants felt in control of decisions

This notion is further reinforced by figure 22 which displays individuals felt in control of their decisions which indicates that the game flow is understandable and smooth, even to players that have no experience in gaming.

5.5.2 – Verbal feedback and perceptions

Participants offered the following reflections:

- “The enemy’s AI was fair and made sense, however the game ended too quickly”
- “I didn’t fully understand the elemental system until halfway through the game”
- “Defend was useful but I didn’t find a chance to use it under stress”
- “I preferred using heal over defend”
- “The enemy was defeated too fast”
- “I liked that I could use weak moves and still win if I adapted my play style”

These responses demonstrate emotional immersion and strategic tension which validates the game’s design principles which were discussed in the third chapter. The main takeaway from these qualitative responses is that the enemy is too quick to beat and that some mechanics could be perceived as confusing

Additionally, the heal mechanic was preferred over the defend mechanic despite the heal mechanic’s penalty system. Players claimed that this is because defending was only a temporary measure while healing seemed permanent and long-term.

5.5.3 – Implications and improvements

Based on pilot feedback, it has been decided a tutorial and explanation system for mechanics can be implemented before the game begins to explain mechanics to players. Increasing the enemy’s health would also let the game go on for longer and help the players feel an increased sense of challenge

5.5.2 – Triangulation of data

By matching player IDs from the google forms to player IDs produced by the game when gameplay is finished, some interesting trends can be found from the 10 players that did participate in the experiment.

The first example is a player who claimed that they prefer to use low-risk strategies. This same player also claimed that they avoided using elemental attacks due to the risk associated with them. When asked about their decisions in the open-ended questions, their responses supported their decisions made in the game as they claimed that they chose to avoid elemental attacks because of the risk and instead tried to heal more when close to loss. When cross-referenced to gameplay, this notion is supported

	A	B	C	D	E	F	G	H	I
1	PlayerID	MoveNum	Role	MoveType	MoveTier	Element	Outcome	Value	Success
2	DF3GCSZ7	1	Player	Attack	Low Attack	Neutral	Hit for 15 damage	15	TRUE
3	DF3GCSZ7	2	Player	Attack	Low Attack	Neutral	Hit for 15 damage	15	TRUE
4	DF3GCSZ7	3	Player	Attack	Low Attack	Neutral	Hit for 15 damage	15	TRUE
5	DF3GCSZ7	4	Player	Heal	Low Heal	n/a	Healed for 10	10	TRUE
6	DF3GCSZ7	5	Player	Heal	High Heal	n/a	Healed for 40	40	TRUE
7	DF3GCSZ7	6	Player	Attack	Low Attack	Neutral	Hit for 15 damage	15	TRUE
8	DF3GCSZ7	7	Player	Heal	Mid Heal	n/a	Healed for 25	25	TRUE
9	DF3GCSZ7	8	Player	Attack	Low Attack	Neutral	Hit for 15 damage	15	TRUE
10	DF3GCSZ7	9	Player	Attack	Mid Attack	Neutral	Hit for 30 damage	30	TRUE
11	DF3GCSZ7	10	Player	Heal	High Heal	n/a	Heal failed	0	FALSE
12	DF3GCSZ7	11	Player	Heal	High Heal	n/a	Healed for 40	40	TRUE
13	DF3GCSZ7	12	Player	Heal	Low Heal	n/a	Healed for 10	10	TRUE
14	DF3GCSZ7	13	Player	Heal	High Heal	n/a	Heal failed	0	FALSE
15	DF3GCSZ7	14	Player	Attack	Berserk Strik	Neutral	Missed	0	FALSE

Figure 23 - A single player's gameplay data

The player does indeed prefer to use low-risk attacks most of the time and only chooses higher risk attacks when their health gets lower or when they get close to losing

Chapter 6 – Conclusion

This project set out to design and develop an interact game than could serve as a novel tool for measuring risk-taking behaviour which is grounded in established psychological theory. Through the integration of core gameplay mechanics, optional elemental volatility and desperation driven choices, the game successfully models different forms of calculated, affective and strategic risk-taking

From a mechanical perspective, the game offers the players with meaningful autonomy, emotional tension, and a variety of behavioural pathways, while also maintaining structured data collection. The adaptive AI system, simultaneous move resolution system, and optional Berserk mode were especially effective in enhancing realism and preserving experimental

validity. The CSV logging and questionnaire design enable mixed-methods analysis, supporting both behavioural and motivational interpretations of player decision-making.

Despite all these strengths, there are several limitations. The AI's adaptiveness remains rule based rather than truly predictive, and the sample size for analysis is likely to be small, limiting generalisability. Additionally, the game currently only supports single-player data could lack long-term behavioural tracking.

Regardless of this, the project has demonstrated the feasibility and value of interactive, game-based tools for psychological research. This project contributes to the growing field of gamified experimentation by offering a scalable and engaging alternative to linear tasks like the BART or IGT.

Future work could potentially focus on expanding the complexity of the AI, incorporating multiplayer comparisons and refining feedback mechanisms to track player learning over time. With these enhancements, the system has strong potential to evolve into a fully deployable behavioural testing tool for academic and clinical testing/research.

References

- Bechara, A., Damasio, A.R., Damasio, H. and Anderson, S.W. (1994) 'Insensitivity to future consequences following damage to human prefrontal cortex', *Cognition*, 50(1–3), pp. 7–15. Available at: [https://doi.org/10.1016/0010-0277\(94\)90018-3](https://doi.org/10.1016/0010-0277(94)90018-3)
- Burdick, J.D., Roy, A.L. and Raver, C.C. (2013) 'Evaluating the Iowa Gambling Task as a direct assessment of impulsivity with low-income children', *Personality and Individual Differences*, 55(7), pp. 771–776.
- Casey, B.J. and Jones, R.M. (2010) 'Neurobiology of the adolescent brain and behavior: Implications for substance use disorders', *Journal of the American Academy of Child & Adolescent Psychiatry*, 49(12), pp. 1189–1201. Available at: <https://doi.org/10.1016/j.jaac.2010.08.017>
- Clark, L., Lawrence, A.J., Astley-Jones, F. and Gray, N. (2009) 'Gambling near-misses enhance motivation to gamble and recruit win-related brain circuitry', *Neuron*, 61(3), pp. 481–490. Available at: <https://doi.org/10.1016/j.neuron.2008.12.031>
- Cyders, M.A. and Coskunpinar, A. (2011) 'Measurement of constructs using self-report and behavioral lab tasks: Is there overlap?', *Psychological Assessment*, 23(3), pp. 620–627.
- Habib, R. and Dixon, M.R. (2010) 'Neurobehavioral evidence for the “near-miss” effect in pathological gamblers', *Journal of the Experimental Analysis of Behavior*, 93(3), pp. 313–328. Available at: <https://doi.org/10.1901/jeab.2010.93-313>
- Kahneman, D. and Tversky, A. (1979) 'Prospect theory: An analysis of decision under risk', *Econometrica*, 47(2), pp. 263–291. Available at: <https://www.jstor.org/stable/1914185>
- Knežević, S. and Jovanović, M. (2021) 'Simulating a questionnaire on the framing effects in decision-making processes in a serious game', *arXiv preprint*. Available at: <https://arxiv.org/pdf/2110.15011.pdf>
- Langer, E.J. (1975) 'The illusion of control', *Journal of Personality and Social Psychology*, 32(2), pp. 311–328. Available at: <https://doi.org/10.1037/0022-3514.32.2.311>
- Lejuez, C.W. et al. (2002) 'Evaluation of a behavioral measure of risk taking: the Balloon Analogue Risk Task (BART)', *Journal of Experimental Psychology: Applied*, 8(2), pp. 75–84. Available at: <https://doi.org/10.1037/1076-898X.8.2.75>
- Steinberg, L. (2010) 'A dual systems model of adolescent risk-taking', *Developmental Review*, 30(1), pp. 56–79. Available at: <https://doi.org/10.1016/j.dr.2009.07.003>
- Thompson, S.C., Armstrong, W. and Thomas, C. (1998) 'Illusions of control, underestimations, and accuracy: A control heuristic explanation', *Psychological Bulletin*, 123(2), pp. 143–161. Available at: <https://doi.org/10.1037/0033-2909.123.2.143>

- Weber, E.U. and Hsee, C. (1998) 'Cross-cultural differences in risk perception, but cross-cultural similarities in attitudes towards perceived risk', *Management Science*, 44(9), pp. 1205–1217. Available at: <https://doi.org/10.1287/mnsc.44.9.1205>
- Zuckerman, M. (2007) *Sensation Seeking and Risky Behavior*. Washington, DC: American Psychological Association.
- Hamari, J., Koivisto, J. and Sarsa, H. (2014) 'Does gamification work? — A literature review of empirical studies on gamification', *Proceedings of the 47th Hawaii International Conference on System Sciences*, pp. 3025–3034. Available at: <https://doi.org/10.1109/HICSS.2014.377>
- Fernie, G., Cole, J.C., Goudie, A.J. and Field, M. (2010) 'Risk-taking but not response inhibition or delay discounting predict alcohol consumption in social drinkers', *Drug and Alcohol Dependence*, 112(1–2), pp. 54–61. Available at: <https://doi.org/10.1016/j.drugalcdep.2010.05.011>
- Hunt, M.K., Hopko, D.R. and Bare, R.L. (2005) 'Behavioral assessment of risky sexual behavior in college students using the Balloon Analogue Risk Task', *The Journal of Sex Research*, 42(3), pp. 259–267. Available at: <https://doi.org/10.1080/00224490509552282>
- Reynolds, E.K., Ortengren, A., Richards, J.B. and de Wit, H. (2006) 'Dimensions of impulsive behavior: Personality and behavioral measures', *Personality and Individual Differences*, 40(2), pp. 305–315. Available at: <https://doi.org/10.1016/j.paid.2005.03.024>
- Pleskac, T.J., Wallsten, T.S., Wang, P. and Lejuez, C.W. (2008) 'Development of an automatic response mode to improve the clinical utility of the Balloon Analogue Risk Task (BART)', *Psychological Assessment*, 20(3), pp. 253–258. Available at: <https://doi.org/10.1037/1040-3590.20.3.253>
- White, T.L., Lejuez, C.W. and de Wit, H. (2008) 'Test-retest characteristics of the Balloon Analogue Risk Task (BART)', *Experimental and Clinical Psychopharmacology*, 16(6), pp. 565–570. Available at: <https://doi.org/10.1037/a0014083>
- Wallsten, T.S., Pleskac, T.J. and Lejuez, C.W. (2005) 'Modeling behavior in a clinically diagnostic sequential risk-taking task', *Psychological Review*, 112(4), pp. 862–880. Available at: <https://doi.org/10.1037/0033-295X.112.4.862>
- Figner, B., Mackinlay, R.J., Wilkening, F. and Weber, E.U. (2009) 'Affective and deliberative processes in risky choice: Age differences in risk taking in the Columbia Card Task', *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 35(3), pp. 709–730. Available at: <https://doi.org/10.1037/a0014983>
- Gescheidt, T., Czekoova, K., Urbanek, T., et al. (2012) 'Iowa Gambling Task in patients with early-onset Parkinson's disease: Strategy analysis', *Neuropsychologia*, 50(2), pp. 321–329. Available at: <https://doi.org/10.1016/j.neuropsychologia.2011.11.019>

Buelow, M.T. and Suhr, J.A. (2009) 'Construct validity of the Iowa Gambling Task', *Neuropsychology Review*, 19(1), pp. 102–114. Available at: <https://doi.org/10.1007/s11065-009-9083-4>

van den Bos, R., Homberg, J. and de Visser, L. (2013) 'A critical review of sex differences in decision-making tasks: Focus on the Iowa Gambling Task', *Behavioural Brain Research*, 238, pp. 95–108. Available at: <https://doi.org/10.1016/j.bbr.2012.10.002>

Dunn, B.D., Dalgleish, T. and Lawrence, A.D. (2006) 'The somatic marker hypothesis: A critical evaluation', *Neuroscience & Biobehavioral Reviews*, 30(2), pp. 239–271. Available at: <https://doi.org/10.1016/j.neubiorev.2005.07.001>

Chiu, Y.C., Lin, C.H., Huang, J.T., Lin, S. and Lee, P.L. (2008) 'Immediate gain is long-term loss: Are there foresighted decision makers in the Iowa Gambling Task?', *Behavioral and Brain Functions*, 4(1), p. 13. Available at: <https://doi.org/10.1186/1744-9081-4-13>

Maia, T.V. and McClelland, J.L. (2004) 'A reexamination of the evidence for the somatic marker hypothesis: What participants really know in the Iowa gambling task', *Proceedings of the National Academy of Sciences*, 101(45), pp. 16075–16080. Available at: <https://doi.org/10.1073/pnas.0406666101>

Chiu, Y.C. and Lin, C.H. (2007) 'Is deck B a disadvantageous deck in the Iowa Gambling Task?', *Behavioral and Brain Functions*, 3(1), p. 16. Available at: <https://doi.org/10.1186/1744-9081-3-16>

Bussemeyer, J.R. and Stout, J.C. (2002) 'A connectionist model of decision-making in dynamic environments', *Psychological Review*, 109(3), pp. 606–621. Available at: <https://doi.org/10.1037/0033-295X.109.3.606>

Buelow, M.T. and Blaine, A.L. (2015) 'The assessment of risky decision making: A comparison of performance on the Iowa Gambling Task and the Balloon Analogue Risk Task', *Journal of Clinical and Experimental Neuropsychology*, 37(10), pp. 1025–1034. Available at: <https://doi.org/10.1080/13803395.2015.1067290>.

van Duijvenvoorde, A.C.K., Huizenga, H.M., Somerville, L.H., Delgado, M.R., Powers, K.E., Weeda, W.D., Casey, B.J. and Figner, B. (2015) 'Neural correlates of expected risks and returns in risky choice across development', *Journal of Neuroscience*, 35(4), pp. 1549–1560. Available at: <https://doi.org/10.1523/JNEUROSCI.1924-14.2015>.

Landers, R.N. (2014) 'Developing a theory of gamified learning: Linking serious games and gamification of learning', *Simulation & Gaming*, 45(6), pp. 752–768. Available at: <https://doi.org/10.1177/1046878114563660>.

Deci, E.L., Koestner, R., & Ryan, R.M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin*, 125(6), 627–668.

Creswell, J.W. & Plano Clark, V.L. (2011). *Designing and Conducting Mixed Methods Research* (2nd ed.). SAGE Publications.

Keetharuth, A.D., Brazier, J., Connell, J., Barkham, M., & Lovell, K. (2018). Integrating qualitative and quantitative data in the development of outcome measures: The case of the recovering quality of life (ReQoL) measures in mental health populations. *International Journal of Environmental Research and Public Health*, 15(6), 1149.

<https://doi.org/10.3390/ijerph15061149>

Teddlie, C. & Tashakkori, A. (2010). Overview of contemporary issues in mixed methods research. In A. Tashakkori & C. Teddlie (Eds.), *SAGE Handbook of Mixed Methods in Social & Behavioral Research* (2nd ed., pp. 1-41). SAGE Publications.

Fetters, M.D., Curry, L.A., & Creswell, J.W. (2013). Achieving integration in mixed methods designs—principles and practices. *Health Services Research*, 48(6pt2), 2134-2156.

<https://doi.org/10.1111/1475-6773.12117>

Schell, J. (2008). *The Art of Game Design: A Book of Lenses*. CRC Press.

Nielsen, J. (1995). *10 usability heuristics for user interface design*. Nielsen Norman Group.

<https://www.nngroup.com/articles/ten-usability-heuristics/>

Friedman, B., & Hendry, D. G. (2019). *Value Sensitive Design: Shaping Technology with Moral Imagination*. MIT Press.

Norman, D. (2013). *The Design of Everyday Things*. MIT Press.

WAI – Web Accessibility Initiative. (2023). <https://www.w3.org/WAI/>

Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N., & Diakopoulos, N. (2010). *Designing the User Interface: Strategies for Effective Human-Computer Interaction*. Pearson.

Iacovides, I. et al. (2015). "Removing the HUD: The Impact of Non-Diegetic Game Elements on Player Involvement." *CHI Play 2015*

Csikszentmihalyi, M. (1990) *Flow: The Psychology of Optimal Experience*. New York: Harper & Row.

BPS (British Psychological Society), 2021. *Ethical guidelines for psychological research online*. Leicester: British Psychological Society. Available at: <https://www.bps.org.uk/guidelines-and-resources> [Accessed 19 May 2025].

American Psychological Association (APA), 2017. *Ethical principles of psychologists and code of conduct*. Washington, D.C.: APA. Available at: <https://www.apa.org/ethics/code> [Accessed 19 May 2025].

Figner, B. and Weber, E.U., 2011. Who takes risks when and why? Determinants of risk-taking. *Current Directions in Psychological Science*, 20(4), pp.211–216. Available at: <https://doi.org/10.1177/0963721411415790> [Accessed 19 May 2025].

Lehmann, M., Häusser, J.A. and Mojzisch, A., 2020. The effects of data collection methods on psychological research outcomes. *Frontiers in Psychology*, 11, p.621. Available at: <https://doi.org/10.3389/fpsyg.2020.00621> [Accessed 19 May 2025].

Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), pp.77–101. Available at: <https://doi.org/10.1191/1478088706qp063oa> [Accessed 19 May 2025].

Appendices

Appendix 1 – Participant information sheet

Participant Information Sheet

Before you decide whether to take part, it is important that you understand what you will be required to do. Please take the time to read the following information and feel free to ask if anything is not clear or if you need more information.

What is the purpose of the study?

In this study, we would like you to try to use the web-game I have made and then fill in a questionnaire.

Do I have to take part?

No. It is up to you to decide whether or not to take part. If you do decide to take part, you will be asked to sign a consent form. However, you will still be free to withdraw at any time and without giving a reason.

What will happen to me if I take part?

You will be invited to use the web-game I have made and then fill in a questionnaire according to the instructions that I give you.

Your responses will be recorded during the experiment. You will be able to communicate with the experimenter during the experiment if you need to do so. If you are uncomfortable or have a problem, it is possible to take a break or stop the experiment altogether at any time.

Once the study is completed, you will be debriefed, and may ask any questions about the study or the procedure.

What are the possible disadvantages and risks of taking part?

The risk associated with this study is that it may cause psychological distress or harm depending on the individual.

Will my taking part in this study be kept confidential?

Yes. All information will be treated in strict confidence. Data will be stored on password protected computers and only the research team will have access to the data. Your test results will not be revealed to anyone, and will be stored anonymised and separate from any personal data like your name.

What will happen to the results of the research study?

The findings of this study will be used in scientific publications and conferences. However, we will keep your name and personal details confidential, and we will ensure that you cannot be identified in these reports.

Please note that new data protection legislation has come into effect across the UK; this means that we need to provide you with some further information relating to how your personal information will be used and managed within this research project.

Manchester Metropolitan University will act as the Data Controller for this study. This means that the University is responsible for looking after your information and using it properly. In order to collect and use your personal information as part of this research project, we must have a basis in law to do so. The basis that we are using is that the research is 'a task in the public interest'.

Further information, including details about how and why the University processes your personal information, how we keep your information secure, and your legal rights (including

how to complain if you feel that your personal information has not been handled correctly), can be found here: <https://www.mmu.ac.uk/data-protection/dppolicy>.

Contact for further information

If you require any further information, please contact *Ahsan Siddiqui*, Department of Computing and Mathematics, email: 24489934@stu.mmu.ac.uk. Alternatively, you may contact *Stephen Gordon*, Department of Computing and Mathematics, email: s.gordon@mmu.ac.uk.

Appendix 2 – Participant consent form

PARTICIPANT CONSENT FORM

Title of Study: *Risk-taking psychology web game*

Lead researcher: *Ahsan Siddiqui* (22489934@stu.mmu.ac.uk)

Participant Identification Number for this project:

	Please initial
1. I confirm that I have read and understood the information sheet explaining the above research project and I have had the opportunity to ask questions about the project.	
2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason and without any negative consequences. In addition, should I not wish to answer any particular question or questions, I am free to decline.	
3. I am happy for my responses to be analysed.	
4. I understand that my responses will be anonymous. I give permission for members of the research team to have access to my anonymised responses. I understand that my name will not be linked with the research materials, and I will not be identified or identifiable in the report or reports that result from the research.	
5. I understand that my taking part is voluntary and that I can withdraw from the study at any time before 22/05/2025; I do not have to give any reasons for why I no longer want to take part and there will be no adverse consequences if I choose to withdraw.	
6. I agree to take part in the above research project. I understand that taking part in the project will involve completing the web-game tasks and a questionnaire	
How my information will be used during and after the project	
I understand my personal details such as name, phone number, address and email address etc. will not be revealed to people outside the project.	
I understand and agree that my words may be quoted in publications, reports, web pages, and other research outputs. I understand that I will not be named in these outputs unless I specifically request this.	

I understand and agree that other authorised researchers will have access to this data only if they agree to preserve the confidentiality of the information as requested in this form.

I understand and agree that other authorised researchers may use my data in publications, reports, web pages, and other research outputs, only if they agree to preserve the confidentiality of the information as requested in this form.

I give permission for the responses in the estimating and gambling tasks, and the questionnaire conducted during the experiment, and data relating to my eye movements to be deposited in One Drive so it can be used for future research and learning.

So that the information you provide can be used legally by the researchers

I agree to assign the copyright I hold in any materials generated as part of this project to Manchester Metropolitan University

Personal Details

Please give an email address (so that we can contact you should you win):

Please indicate your age

Are you Male / Female / other / prefer not to say

Name of Participant Date Signature

Lead Researcher Date Signature

To be signed and dated in presence of the participant.